REFERENCES

No	Document	Description
1	5.2.2.1 - Site Specific Information_Rev 1	Owner Growth Project - Process Plant - Site Specific Information
2	4.7.1 - Consultant Report 7555 - Owner PDC Data Summary Rev 0, March 2022	Table 2-5, "Back Calculated Ore Parameters", 2014 - 2021 value.
4	4.6.1 - Owner Growth Mass Balance_Nominal Case Report_Rev C	Mass Balance - SysCAD Report Nominal Case. 22/03/2022
5	Owner Comminution Plant Data_B.xlsx"	Benchmarking from FIM pebble crushing circuit.
7	ECI Tender from Consep	Q12375 - EPC Gravity Circuit Rev1
8	5.1.12f - BOD Alignment Template - Flotation Tail PLT, FT CIL, Elution & EW	Basis of Design for Flotation Tailings CIL, Elution & EW circuit. 06/04/2022
9	4.7.3 - Consultant Report 7555 Owner PDC PSD Addendum Rev 1	Owner - Process Design Criteria Summary - PSD Addendum Report No. 7555 Rev 1 March 2022
10	5.1.12g - BOD Alignment Template - Reagents and Services	5.1.12g - BOD Alignment Template - Reagents and Services - Rev C Owner confirmation that regrind mill ball size will be 25 mm (cell F15).
11	EPC Calculation CAL-PR-002	Flotation Tailings CIL Simulation @ 8 and 10 g/L Carbon Concentration
14	Owner Growth Project ECI Outcome Letter - EPC	Owner Growth Project - Process Plant - Formal Agreement for Engineering, Procurement and Construction: Early Contractor Involvement (Value Engineering Phase). 03/08/2022
15	5.1.12b - BOD Alignment Template - 13 Grinding - Rev C	5.1.12b - BOD Alignment Template -13 Grinding - Rev C - EPC (220609)_NSR Review
16	Owner Process Design Criteria - VEP Phase (Stage 1)_I	Owner - Growth Project - Process Design Criteria Rev I. 20/09/2022
18	Cleaner Option 2- Owner Case 2 Scav Con to Cleaner	Glencore Technology - Jameson Cell Mass Balance - Cleaner
19	Cleaner Scavenger - Owner Case 2 Scav to Cleaner	Glencore Technology - Jameson Cell Mass Balance - Cleaner Scavenger
23	Owner GP Process Design Criteria - Reliance Information S1 Rev K, 25/01/2023	Principal supplied reliance information.
24	VEP Outcome Letters - EPC 09/02/23 and 020523	Principal supplied Scope of Works letter for Stage 1 and 2
25	Consultant Report 7653 - Owner Value Engineering Update	Grinding circuit modelling for Owner by Orway Mineral Consultants, 160223
26	Process Design Workshop Meeting Minutes 170223, MOM-PM-SFP002	Workshop between the Client and EPC
27	Glencore Jameson Cell Sizings and Quote	Simulations from Vendor, 22/02/24 Owner Expansion Project - Jameson Cell Quote OTJ-2220 7532.3
28	Weir Cyclone Simulations and Datasheet - Primary & Gravity Cyclones	Various simulations for the different process cases, and datasheet: 2000-F00-DSH-ME-11005A_B Primary Cyclone
31	Trash Screen Spray Water Information from Vendor	Emails with Vendor, 16/08/23
32	Flotation Mass Balance Scenarios	Mass Balances for Process Margins Rev H and New Mass Balance Cases Rev G
33	Cadia Gravity Electrowinning Cell	Emails with Vendor, 14/03/23. Ag and Cu Assays from site, provided by Owner (23/03/23)
35	Discussions with Client and TQ regarding Cyanide Destruction	Emails with Client, 230323 and TQ: 2000-F72-TQY-ME-10094_0_TQ-R
36	Vendor Information from Transmin for New Flocculant Mixing Plant	Transmin Proposal Execution Phase rev 3, P33-904-06
37	Direction from EPC Mechanical Engineers re: COS	Email, 280323
38	Client Information regarding Grinding Media	Emails with Client, 070323
39	Vendor Information from Consep for the Acacia CS8000 ILR	Emails with Client, 210322
40	EPC Calculation	"SAG_Ball_Screen_Trommel_Calc_Rev_A.xlsx"
41	Client Direction, Transfer of FLS Equipment Technical Information	Emails from client, various dates (May 2023)
42	Primary Crusher Datasheet	Scope of Work - Primary Crusher, Appendix A, doc 2000-F12-SOW-ME-20001 rev 1
43	SoW for Separable Portions Stage 1&2	Scope of Works between Owner and EPC 2000-F00-SOW-CO-20001 & 2000-F00-SOW-CO-20002_0
44	Outstanding Process Inputs from Owner	Outstanding Process Inputs 210623 (version 1) Additional (003) (002), and 2000 F00-TQY-PR-10051.0.TQ-R.0.01
47	Oxygen Consumption Information	Emails from client, 8/5/23
48	Cadia CIL4 Electrowinning Cells	Vendor Quotation 27/03/2024
49	Owner email Owner GP - Float Cell Motors", dated 11 May 2023	FLSmidth flotation blower details.
50	Calculation for LPG Consumption and TPE ECI Tender	CAL-PR-005_A LPG Consumption TPE5000 ELUTION HEATER PACKAGE - ME16 TENDER - COMPLETE 20/05/22
51	FLSmidth - Primary Crusher Application Model	FLS_13x19 TSUV_Calc_FIM_Run 15_13x19 TSUV_ 6.5 OSS_3716 mtph_70, AVG results
52	Vendor Information Mercury Abatement System (CIL4 Carbon Regen Kiln) and Client Direction Variation 25; Vendor information for Lochhead CIL4 Carbon Regen Kiln; Regen Kiln Off Gas Calculations	Information from the Mercury Abatement Vendor, Jan 2024 P3-EPC-COR-161 Variation Direction No.025 Carbon regeneration kiln off-gas mercury abatement system Lochhead Carbon Reactivation Kiln (ME-17) Quotation No. 5211 (August 26, 2022) Kiln Off Gas Calc
53	EPC email Apron Feeder Datasheet Internal Review", dated 30 August 2023	Agreement with Owner to change contract SOW design factor from 120% to 110% of nominal tonnage rate.
54	Weir UFG Cyclone Cluster Modelling	UFG_CY_Select_Design_RevC UFG_CY_Select_Nominal_RevC
55	Takraf Tender Rev 0 Datasheet	2000-F00-DSH-ME-11018_B Trash Screens Completed
56	Owner GP Process Design Criteria - Reliance Information S1 Rev L, 25/01/2023	Principal supplied reliance information.
57	Owner Growth Project SysCAD Model (Historical)	SysCAD Model files and Reports (2022.08.31)-01.spf
58	NTF Deslime & UFG Cyclone PDC Inputs	Word document 260723
59	Joest Vibrating Screens Offer	JOEST_M22-11102 Rev D_JOEST FIRM Offer 2000-F00-DSH-ME-11011_C Vibrating Screens
60	Flotation Launder Wash Water Technical Query	Technical Query (2000-F16-TQY-ME-10007) FLS Launder Water Requirements Rev.0 31/07/2023
62	Technical Query - Design Bulk Density Range for Apron Feeders And Conveyors	2000-F00-TQY-ME-10034

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No	Document	Description
63	Flotation Area - Design P80 Sizes	Email Correspondence. RE - Flotation D80 Mech Requests. 250823
64	Reagent Delivery Formats, 2000-F72-TQY-ME-20001_2	Reagent delivery information, provided by Owner
65	IsaMill Quote and Modelling by Glencore	IsaMill Quote OMT232007747 Owner - Owner EPC - IsaMill Sizing - Design -19102023 - New data supplied water temp 25 Owner - Owner EPC - IsaMill Sizing - Nominal -19102023 - New data supplied water temp 25 Email: "FW_ ME039 IsaMill Latest MB - Clarifications 64 - UFG Mill Media Hopper Live Volume"
66	Regrind Removal TQ	2000-F00-TQY-GE-10080_0 - RESPONSE.pdf
67	Client Comments Stage 1 Rev K PDC and Rev L SysCAD	2000-F00-DCR-PR-10001.K.ICR-R.K.02
68	Raw Water TQ 10044	2000-F24-TQY-PR-10044.0.TQ-R.0.0164
69	SysCAD & PDC Client Comments TQ 10107 Response	2000-F00-TQY-PR-10107.0.TQ-R.0.02
70	Gravity Variation (VAR-017) and Communication from the Client	P3-EPC-COR-085 Variation Direction No.017 Gravity Circuit Modifications Emails from Client/Outcome of meetings
71	Emails from Consep (Knelson & ILR) in relation to Gravity Variation (VAR-017)	Email dated 30/10/23, 'RE: Knelson Concentrators (KC-QS70) and CS10000'
72	Gravity Variation TQ 10115 Response from Client	2000-F14-TQY-PR-10115.0.TQ-R.0.01 2000-F14-TQY-PR-10115_1_TQ-R
73	Quotation and emails from Cadia Systems (EW cells) in relation to Gravity Variation (VAR-017)	Q231115-PRI_EPC_EWC 151123 issue RE_ Owner - Gravity Pregnant Solution Electrowinning - various emails
74	Variation 018 UFG Mill 2 Trash Screen	P3-EPC-COR-104 Variation Direction No.018 UFG Mill 2 Trash Screen
75	Clarification Emails for Flotation Reagent Calculations	Emails with Client 21/11/23: 'RE: Owner GP - PAX Usage/Flow Consumption/Storage Tank RT'
76	Dump Pocket TQ from Owner	2000-F12-TQY-GE-10127_0_TQ-R
77	Wet Static Screens Datasheet (ME40) and Spray Water Information	2000-F00-DSH-ME-11040_1 and emails dated 23/01/24
78	Owner Owner's Manual CAT793F Owner	Owner's Manual CAT793F Owner
79	Water Services PFD Client Review	2000-F00-PFD-PR-10046.D.IFD-R.D.02
80	Area 30 M1 Design Review Report	2000-F30-REP-GE-10101_A - M1 Design Review Report
81	TPE CIL4 Elution Heater Vendor Package	TPE5000 ELUTION HEATER PACKAGE_DES.pdf rev A 20/05/22
82	Client Information LPG Storage Vessel	Email: RE LPG Tank, 22/05/2023
83	TQ147 Reagent Tank Volumes, Var 26 Cyanide Tanks and Var 192 CuSO4 Tanks	2000-F72-TQY-ME-10147_0_TQ-R P3-EPC-COR-162 Variation Direction No.026 Cyanide Tank Storage Capacity P3-EPC-COR-192 Variation Direction No.030 Copper Sulphate Storage
84	Joest Tender Documents	JOEST_M22-11102 Rev H_JOEST FIRM Offer, 06/05/24 RE Growth Project - Stage 2 Vibrating Screen Changes
85	Gekko Carbon Scout ECI Tender Documents	6454-EPC-Carbon Scout-220426, 22/04/22
86	ME65 Carbon Transfer Pump Vendor Information	Weir Budget Quote WMFN20220527
87	ME07 Intertank Screens Vendor Information	P28337_Northern Star_EPC Group Growth (Aus) REV 2
88	TQ206 Request for LPG & RTO Information	2000-F00-TQY-PR-10206.0.TQ-R.0.0270
89	2000-F00-CAL-ME-F23TK404 Final Tails Tank Live Vol & OF Calc.xlsx	2000-F00-CAL-ME-F23TK404 Final Tails Tank Live Vol & OF Calc.xlsx
90	Client Comments Stage 1 Rev 0 PDC	2000-F00-DCR-PR-10001.0.IFD-R.0.02
91	Stage 2 Oxygen Consumption Calculations	CAL-PR-003 S2 Oxygen Consumption Calculations
92	SAG & BM Trommel Sizes	2000-ME001.2-B02-50041_C SAG Mill Trommel Sizing Report 2000-ME001.2-M03-50001_B SAG Mill Trommel Assembly 2000-ME001.3-B02-50020_B Ball Mill Trommel Sizing Report 2000-ME001.3-M03-50001_B Ball Mill Trommel Assembly
100	5.1.1 - ECS Process Design Criteria	Enginee Process Design Criteria - Owner Growth Project ECS. 25/01/2022 J6027-000-DCR-PR-001 Rev D
101	5.1.2 - ECS Mass Balance	Enginee Mass Balance - Owner Growth Project ECS. 25/01/2022 J6027-000-CAL-PR-001 Rev D
102	4.5.1 - Process Design Criteria Rev F	Owner - Growth Project - Process Design Criteria Rev F. 16/03/2022
103	4.5.1 - Owner Process Design Criteria - VEP Phase _H, 19/08/2022	Principal supplied reliance information.
104	Email "Carbon Safety Screen" dated 16/02/2024	Confirmation from Owner that currently proposed screen sizes for Carbon Safety Screens is adequate
105	2000-F30-TQY-PR-15001.0.TQ-R.0.02	Client Response to Technical Query - CIL2/3 - Carbon Transfer Rate Between Adsorption Tanks
106	2000-F00-DCR-PR-10002.C.ICR.-R.C.02	Client comments to Process Design Criteria - Stage 2
107	2000-F72-TQY-ME-10260_0-R	Clients Response to Technical Query - Milk of Lime Tank Volume & Residence Time
108	2000-F21-TQY-PR-15014.0.TQ-R.0.02	Clients Response to Technical Query - CIL2/3 & CIL4 Target Process Data
109	2000-F00-TQY-PR-15020.0.TQ-R.0.02	Clients Response to Technical Query - CIL4 Carbon Advance
110	2000-F22-TQY-PR-15018_1_RSP	Clients Response to Technical Query - HCl Dilution Tank
111	Hydrogen Peroxide Technical Data Sheet	Solvay Hydrogen Peroxide Technical Datasheet

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No	Document	Description
112	PAX, CuSO4 & Lead Nitrate Delivery Sizes	Email from IXOM (vendor), dated 28/03/2024
113	Concentrate Thickener Vendor GA	2000-ME091-B02-10001_C
114	Client Comments Stage 2 Rev 0 PDC	20000-F00-DCR-PR-10002_0_AAN
115	2000-F30-TQY-PR-15029_0_TQ-R	Clients Response to Technical Query - Concentrate Pre-Leach Thickener Underflow
116	2000-F72-TQY-PR-15033_0_TQ-R	Clients Response to Technical Query - Lime System Process Design
117	QUO-07322-P7S5G0	Derrick quotation for Derrick Linear Motion Screening Machine 15/08/2024
118	BHT Lime Slaking Package	2000-F00-DSH-ME-12013_0 2000-MEP-ME013-REG_A 706-PC-01 706-PC-02 FW_Lime Mill Media Consumption.msg
119	2000-F30-TQY-PR-15097_1	CIL3 Carbon Movement TQ

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um					
1.0	**PRODUCTION SCHEDULE**												
1.1	**Throughput**												
	Annual Throughput		t/a	10,500,000	25,500,000	23,997,254	25,500,000	28,350,000	1	Owner	16, 24, 26 & 32		A
	Annual Operating Days		d/a	365	365	365	365	365	1	Owner	16		A
1.2	**Operating Schedules**												
	Crushing												
	Plant Availability		%	65.0	65.0	65.0	65.0	65.0	1	Owner	1, 30		A
	Annual Operating Hours		h/a	5,694	5,694	5,694	5,694	5,694	6	Calculated			
	Hourly Throughput - Existing Circuit		t/h	1,844	1,844	1,844	1,844	1,844	6	Calculated			
	Hourly Throughput - New Circuit		t/h		2,634	2,370	2,634	3,135	6	Calculated			
	Total Hourly Throughput		t/h	1,844	4,478	4,214	4,478	4,979	6	Calculated			
1.3	**Concentrator**												
	Plant Availability		%	91.3	93.4	93.4	93.4	93.4	1	Owner	16, 23		A
	Annual Operating Hours - Existing SAG Circuit			8,000	-	-	-	-	6	Calculated			
	Annual Operating Hours - New SAG/Ball Mill Circuit		h/a	-	8,182	8,182	8,182	8,182	6	Calculated			
	Hourly Throughput - Existing SAG Circuit		t/h	1,313	1,313	1,313	1,313	1,313	6	Calculated			
	Hourly Throughput - New SAG/Ball Mill Circuit		t/h		1,833	1,650	1,833	2,182	6	Calculated			
	Total Hourly Throughput		t/h	1,313	3,146	2,962	3,146	3,494	6	Calculated			
	UFG Circuit												
	Plant Availability		%	91.3	93.4	93.4	93.4	93.4	1	Owner	16, 23		A
	Annual Operating Hours		h/a	8,000	8,182	8,182	8,182	8,182	6	Calculated			
2.0	**MATERIAL CHARACTERISTICS**												
2.1	**General Ore Properties**												
	Moisture Content		%w/w			1.3			1	Owner	16		A
	Ore Specific Gravity	Nominal				2.90			1	Owner	16		A
		Range				2.70 - 3.10			1	Owner	23		B
	Ore Bulk Density	Normal				1.90			1	Owner	16		A
		Range				1.70 - 2.10			1	Owner	56		B
		Design Range				1.60 - 1.90			1	Owner	62	To be used for all conveyors & apron feeders in the crushing and milling circuits	B
	Angle of Repose		degrees			36			1	Owner	16		A
	Drawdown Angle		degrees			60			1	Owner	16		A
2.2	**Ore Comminution Parameters**												
	Crushing Work Index												
	Testwork Average		kWh/t			17.8			1	Owner	16		A
	Testwork Range		kWh/t			16 - 30			1	Owner	16		A
	Testwork 80th Percentile		kWh/t			18.8			1	Owner	2, 23	2014 - 2021 value (as of rev E)	A
	Abrasion Index												
	Testwork Average					0.145			1	Owner	16		A
	Testwork Range					0.020 - 0.680			1	Owner	16		A
	Testwork 80th Percentile					0.192			1	Owner	16		A
	Ball Mill Work Index	Range	kWh/t			11.5 - 15			1	Owner	41		B
		Design	kWh/t			12.5			1	Owner	2	2014 - 2021 value	A
	A*b	Range				38 - 48			1	Owner	41		B
		Design				43.8			1	Owner	2	2014 - 2021 value	A
2.3	**Pebble Comminution Parameters**												
	Crushing Work Index	Design	kWh/t			18.8			1	Owner	16		A
		Maximum	kWh/t			23.0			1	Owner	23		A
	Bulk Density	Minimum	t/m^3			1.60			1	Owner	16, 62		A
		Maximum	t/m^3			1.90			1	Owner	41, 62		B
2.4	**Nom Ore Grades**												
	Feed Grade - S		% S	1.02	1.02		1.40	1.02	1	Owner	23	Achievable grade back calculated by EPC (Source 32)	A
	Feed Grade - Au		Au (g/t)	1.58	1.58		1.70	1.58	1	Owner	23	Achievable grade back calculated by EPC (Source 32)	A
	Final Tail Grade - S		%	0.18	1.01		1.39	1.01	1	Owner	23	Blue data is from Owner Ref 23	A
	Final Tail Grade - Au		Au (g/t)	0.23	0.23		0.25	0.23	1	Owner	23	Blue data is from Owner Ref 23	A
2.5	**Design Ore Grades**												
	Feed Grade - S		%S	1.49		1.49			1	Owner	23		A
	Feed Grade - Au		Au (g/t)	1.81		1.81			1	Owner	23		A
	Final Tail Grade - S		%	0.29		1.48			1	Owner	23	Blue data is from Owner Ref 23	A
	Final Tail Grade - Au		Au (g/t)	0.38		0.26			1	Owner	23	Blue data is from Owner Ref 23	A
2.6	**Recovery**												
	Total Gold Recovery	Nominal	%		84	84	84	84	6	Calculated			
	Gravity Gold Recovery	Nominal	%		14.00	14.00	14.00	14.00	1	Owner	23	Relative to feed grade	A
	Gravity Gold Recovery	Maximum	%		20.00	20.00	20.00	20.00	1	Owner	23	Relative to feed grade	A
	Flotation Gold Recovery	Nominal	%		71.80	71.80	71.80	71.80	1	Owner	23	Relative to feed grade	A
	Flotation Gold Recovery	Maximum	%		75.00	75.00	75.00	75.00	1	Owner	23	Relative to feed grade	A
	Flotation Sulphur Recovery	Nominal	%		85.10	85.10	85.10	85.10	1	Owner	23	Relative to feed grade	A
	Flotation Sulphur Recovery	Maximum	%		90.00	90.00	90.00	90.00	1	Owner	23	Relative to feed grade	A
	Flotation Concentrate Leach Gold Recovery	Nominal	%		84.00	84.00	84.00	84.00	1	Owner	44	Range 80-88%	A
	Flotation Concentrate Leach Gold Recovery	Maximum	%		88.00	88.00	88.00	88.00	1	Owner	44	Range 80-88%	A
	Flotation Tails Leach Gold Recovery	Nominal	%		34.00	34.00	34.00	34.00	1	Owner	44	34% YTD	A
	Flotation Tails Leach Gold Recovery	Maximum	%		34.00	34.00	34.00	34.00	1	Owner	44	34% YTD	A
2.7	**Gold Production**												
	Nominal		oz Au/a		1,108,133	1,194,627	1,195,298	1,231,759	6	Calculated			
2.8	**Flotation Concentrate**												
	Production		t/a		636,932	875,971	876,099	707,270	6	Calculated			
2.9	**Process Water SG**				1.070	1.070	1.070	1.070	1	Owner	1		A
2.10	**Scheme Water SG**				1.000	1.000	1.000	1.000	1	Owner	1		B
3.0	**CRUSHING CIRCUIT**												
3.1	**Circuit Parameters**												
	Throughput	Nominal	t/h	1,844	2,634	2,370	2,634	3,135	6	Calculated		New circuit	
		Design	t/h		3,135	3,135	3,135	3,135	1	Owner	43		B
		Maximum	t/h		3,436	3,436	3,436	3,436	1	Owner	56	Crusher fall through from FLS and not the circuit design basis	B
	Feed PSD												
	F_{99}		mm		1,000	1,000	1,000	1,000	1	Owner	23		A
	F_{80}		mm		450	450	450	450	1	Owner	23		A
	F_{50}		mm		150	150	150	150	1	Owner	23		A
	F_{20}		mm		25	25	25	25	1	Owner	23		A
	Product PSD												
	P_{80} - Maximum		mm	100	130	130	130	145	1, 5	Owner, Vendor	23, 25		A
3.2	**Dump Pocket**												
	Dump Pocket Type				Direct Tip, Two Sides	Direct Tip, Two Sides	Direct Tip, Two Sides	Direct Tip, Two Sides	1	Owner	16		A
	Truck Model				793F	793F	793F	793F	1	Owner	23		A
	Truck Payload		t		250	250	250	250	1	Owner	78	Owner's Manual CAT793F.	B
	SAE 2:1 Heaped Tray Capacity		m^3		179	179	179	179	1	Owner	76, 78	Owner's Manual CAT793F.	B
	Dump Pocket Capacity		m^3		179	179	179	179	1	Owner	76, 78	Live capacity from one side	B
3.3	**Primary Crusher**												
	Type			Gyratory	Gyratory	Gyratory	Gyratory	Gyratory	1	Owner	16	Owner Supplied	A
	Model			54-75Mk II	1300 x 1900 TSUV	1300 x 1900 TSUV	1300 x 1900 TSUV	1300 x 1900 TSUV	1	Owner	41	Owner Supplied	A
	Open Side Setting		mm	141	165	165	165	165	5	Vendor	51		B
	Throw		mm		38	38	38	38	1	Owner	42		B
	Motor Power		kW	450	600	600	600	600	5	Vendor	42		A
	Simulation Results												
	P_{80}	Max	mm		122	122	122	122	5	Vendor	51	Simulated, see reference document for inputs	B
	P_{100}	Max	mm		229	229	229	229	5	Vendor	51	Simulated, see reference document for inputs	B

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um					
3.4	**Crushed Ore Storage**												
	Type			Stockpile	Stockpile	Stockpile	Stockpile	Stockpile	1	Owner	16		A
	Live Capacity - Existing		t	21,001	-	-	-	-	6	Calculated			
	Residence Time - Existing		h	16	-	-	-	-	6	Calculated			1
	Bulk Density		t/m^3		1.60 - 1.80	1.60 - 1.80	1.60 - 1.80	1.60 - 1.80	1	Owner	43		B
	Live Capacity - New	using 60° draw down	m^3		12,035	12,035	12,035	12,035	2	EPC	37		A
	Residence Time - Minimum/Range	using 60° draw down	h		11.16	12.40	11.16	9.38	6	Calculated			
	Live Capacity at 1.70 t/m^3	using 60° draw down	t		20,460	20,460	20,460	20,460	6	Calculated			
	Live Capacity at 1.90 t/m^3	using 60° draw down	t		22,867	22,867	22,867	22,867	6	Calculated			
	Live Capacity - New	using TUNRA data	m^3		11,293	11,293	11,293	11,293	2	EPC	37		A
	Residence Time - Minimum/Range	using TUNRA data	h		10.47	11.64	10.47	8.80	6	Calculated			
	Live Capacity at 1.70 t/m^3	using TUNRA data	t		19,198	19,198	19,198	19,198	6	Calculated			
	Live Capacity at 1.90 t/m^3	using TUNRA data	t		21,457	21,457	21,457	21,457	6	Calculated			
	Primary Crusher Discharge Feeder												
	Number				1	1	1	1	1	Owner	43		B
	Type				Apron	Apron	Apron	Apron	1	Owner	16		B
	Capacity	Nominal	t/h		2,634	2,370	2,634	3,135	6	Calculated			
		Design	t/h		3,135	3,135	3,135	3,135	1	Owner	43		B
3.5	**Reclaim Feeders**												
	Number			4	2	2	2	2	1, 2	EPC Owner	16, 26		A
	Type			Apron	Apron	Apron	Apron	Apron	1	Owner	16		A
	Capacity per feeder	Nominal	t/h		917	825	917	1,091	6	Calculated			
		Design	t/h		2,017	2,017	2,017	2,017	9	Agreed	53	110% of nominal tonnage rate.	B
3.6	**Emergency Feeders**												
	Number				1	1	1	1	1	Owner	16		A
	Type				Apron	Apron	Apron	Apron	1	Owner	16		A
	Capacity per feeder	Nominal	t/h		1,833	1,650	1,833	2,182	6	Calculated			
		Design	t/h		2,017	2,017	2,017	2,017	9	Agreed	53	110% of nominal tonnage rate.	B
	Loader Model				992K/994K	992K/994K	992K/994K	992K/994K	1	Owner	23		A
4.0	**GRINDING AND CLASSIFICATION**												
4.1	**Circuit Parameters**												
	Type			SABC	SABC	SABC	SABC	SABC	1	Owner	16		A
	Nom Throughput	Nominal @ 180 µm	t/h	1,313	1,833	1,650	1,833		6	Calculated			
	Nom Throughput	Nominal @ 220 µm	t/h					2,182	6	Calculated			
	Bulk Density	Design Range	t/m^3		1.60 - 1.90	1.60 - 1.90	1.60 - 1.90	1.60 - 1.90	1	Owner	43, 62		B
	Milling Circuit PSD												
	F_{80}		mm	100	130	130	130	145	1, 5	Owner, Vendor	23, 25		A
	P_{80}	Target	µm	180	180	180	180	220	1	Owner	16, 23, 24		A
	SAG Mill Specific Energy - 50th Percentile		kWh/t	8.3					1	Owner	16		A
	SAG Mill Specific Energy - 80th Percentile		kWh/t	8.9					1	Owner	16		A
	Ball Mill Specific Energy - 50th Percentile		kWh/t	5.4					1	Owner	16		A
	Ball Mill Specific Energy - 80th Percentile		kWh/t	6.1					1	Owner	16		A
	P_{80}	Modelled	µm		189	189	189	220	1	Owner	25		A
	SAG Mill Specific Energy Modelled		kWh/t		9.14	9.14	9.14	7.68	1	Owner	25		A
	Ball Mill Specific Energy Modelled	@ 180 µm	kWh/t		9.68	9.68	9.68		1	Owner	25		A
	Ball Mill Specific Energy Modelled	@ 220 µm	kWh/t					8.14	2	EPC	25		A
4.2	**Primary Mill**												
	Type			SAG Mill	SAG Mill	SAG Mill	SAG Mill	SAG Mill	1	Owner	16	Owner Supplied	A
	Mill Diameter		m	10.97	10.97	10.97	10.97	10.97	1	Owner	16	Owner Supplied	A
	Effective Grinding Length - EGL		m	4.27	8.38	8.38	8.38	8.38	1	Owner	16, 23	Owner Supplied	A
	Duty Mill Speed Range		% Cs	60 - 80	60 - 83	60 - 83	60 - 83	60 - 83	1	Owner	24		A
	Duty Ball Charge		% v/v	13 - 17	10.5	10.5	10.5	10.9 - 11.3	1	Owner	16, 24		A
	Duty Total Load		% v/v	25 - 30	27	27	27	25	1	Owner	16, 24		A
	Total Ball Charge	Structural	% v/v		18	18	18	18	1	Owner	23	For structural design	A
	Total Load	Structural	% v/v		35	35	35	35	1	Owner	23	For structural design	A
	Mill Discharge Density		% w/w	75	75	75	75	75	1	Owner	9, 69		A
	Mill Pinion Power	Average	kW	10,894	-	-	-	-	1	Owner	25		A
	Mill Pinion Power	Duty	kW	11,682	16,757	15,078	16,757	16,755	6	Calculated			1
	Installed Motor Power		kW	12,000	20,000	20,000	20,000	20,000	1	Owner	41, 23		A
	Liner Type				Steel	Steel	Steel	Steel	1	Owner	41		A
	Curved Pulp Lifter Capacity	Max	m^3/h		2,100	2,100	2,100	2,100	1	Owner	41		B
	Pebble Port Aperture		mm		75	75	75	75	1	Owner	41	FLS offer	A
	Discharge Grate Open Area		%		11 - 12	11 - 12	11 - 12	11 - 12	1	Owner	41		B
	Trommel Dimensions	ID x L	m		3.829 m dia. X 3.80 m	3.829 m dia. X 3.80 m	3.829 m dia. X 3.80 m	3.829 m dia. X 3.80 m	5	Vendor	92		1
	Trommel Open Area		%		25	25	25	25	1	Owner	41		B
	Trommel Aperture		mm		12 x 40	12 x 40	12 x 40	12 x 40	1	Owner	41		B
	Spray Water		m^3/h	150	150	150	150	150	2	EPC	40		B
	Trommel Undersize		t/h		1,050	1,050	1,050	1,050	2	EPC	32		B
	Existing Fim SAG Mill Transfer Density		% w/w	60					1	Owner	57, 69		B
4.3	**Primary Mill Discharge Screen**												
	Discharge Screen Type				Horizontal Vibrating Screen	Horizontal Vibrating Screen	Horizontal Vibrating Screen	Horizontal Vibrating Screen	1	Owner	16		A
	Screen Size	W x L	m		4.27 x 8.54	4.27 x 8.54	4.27 x 8.54	4.27 x 8.54	5	Vendor	59		A
	Effective Screen Area		m^2		36.47	36.47	36.47	36.47	5	Vendor	59		B
	Screen Open Area		%		34.0	34.0	34.0	34.0	5	Vendor	59		B
	Screen Aperture		mm		12 x 55	12 x 55	12 x 55	12 x 55	2	EPC	40		A
	Specific Screen Flux Rate		m^3/m^2/h OA		18.68	18.68	18.68	29.90	5	Vendor	59	Includes wash water	B
	Oversize Moisture		%w/w	5	13	13	13	13	5	Vendor	59		B
	Spray/Wash Water		m^3/h		197.2	197.2	197.2	197.2	5	Vendor	59		B
	Discharge Screen Feed % Solids		%w/w		85	85	85	85	2	EPC	32		B
	Discharge Screen Oversize		t/h		375	375	375	375	1	Owner	41		B
4.4	**Secondary Mill**												
	Type				Overflow Ball Mill	Overflow Ball Mill	Overflow Ball Mill	Overflow Ball Mill	1	Owner	41	Owner Supplied	A
	Mill Diameter		m		8.53	8.53	8.53	8.53	1	Owner	23	Owner Supplied	A
	Effective Grinding Length - EGL		m		13.26	13.26	13.26	13.26	1	Owner	23	Owner Supplied	A
	Duty Mill Speed		% Cs		60 - 78	60 - 78	60 - 78	60 - 78	1	Owner	41		B
	Duty Ball Charge	Nominal	% v/v		28.5	28.5	28.5	28.5	1	Owner	41		A
	Duty Ball Charge	Maximum	% v/v		31.0	31.0	31.0	31.0	1	Owner	24		A
	Total Ball Charge	Structural	% v/v		38	38	38	38	1	Owner	23	With polymet liners. For structural design	A
	Mill Discharge Density		% w/w		75	75	75	75	1	Owner	4		A
	Mill Pinion Power	Duty	kW		17,747	15,969	17,747	17,759	6	Calculated			
	Installed Motor Power		kW		20,000	20,000	20,000	20,000	1	Owner	23		A
	Liner Type				Poly-Met	Poly-Met	Poly-Met	Poly-Met	1	Owner	23		A
	Trommel Flux Rate	Nominal	m^3/m^2/h		230	223	229	226	6	Calculated			
	Trommel Dimensions	ID x L	m x m		4.992 x 5.80	4.992 x 5.80	4.992 x 5.80	4.992 x 5.80	5	Vendor	92	ID = internal diameter	1
	Trommel Capacity	Maximum	m^3/h		6,681	6,681	6,681	6,681	1	Owner	23	Includes spray water	A
	Mill Discharge Flowrate	Nominal	m^3/h		6,373	6,197	6,363	6,277	6	Calculated		Includes spray water	
	Trommel Open Area	Minimum	%		30	30	30	30	1	Owner	23		A
	Trommel Aperture		mm		19 x 40 - slotted	19 x 40 - slotted	19 x 40 - slotted	19 x 40 - slotted	1	Owner	23		A
	Trommel Screen Spray Water		m^3/h		150	150	150	150	1	Owner	4		A
4.5	**Pebble Crusher**												
	Type			Cone	Cone	Cone	Cone	Cone	1	Owner	16	Owner Supplied	A
	Number			2 (duty / standby)	2 (duty / duty)	2 (duty / duty)	2 (duty / duty)	2 (duty / duty)	1	Owner	16, 23	Owner Supplied	A
	Model			Symons 7ft	XL7 SH-HD	XL7 SH-HD	XL7 SH-HD	XL7 SH-HD	1	Owner	41	Owner Supplied	A
	Installed Motor Power per Crusher		kW	375 / 260	375	375	375	375	1	Owner	16, 41, 44		A
	Closed Side Setting	Minimum	mm	7 - 9	7	7	7	7	1	Owner	16, 41		B
	Pebble Crushing Work Index	Maximum	kWh/t		23	23	23	23	1	Owner	56		1
	Pebble Bulk Density	Range	t/m^3		1.6 - 1.9	1.6 - 1.9	1.6 - 1.9	1.6 - 1.9	1	Owner	56		1
	Throughput	Nominal	t/h	302									
	Throughput	Nominal	t/h	302					6	Calculated			
		Nominal	t/h		305	305	305	610	1	Owner	41		B
		Range	% of new feed	23 - 27					1	Owner	16		B
			% of new feed	23.0					1	Owner	16		B
			% of new feed		16.6	18.5	16.6	28.0	6	Calculated			
	Crusher Capacity at CSS	Maximum	t/h		610	610	610	610	1	Owner	41	305 t/h limit per crusher unit at CSS. This throughput represents the increased pebble load as the pebble ports wear out on SAG mill discharge grate	A
	Feed Bin Capacity	per bin	min	50	12.8	12.8	12.8	6	1, 2	Owner, EPC	26		A
			t	250	65	65	65	65	6	Calculated		Total Bin capacity of 130 t (2 bins)	
			m^3	132	34	34	34	34	6	Calculated		Total Bin capacity of 68 m^3 (2 bins)	
	Feed Moisture	Range	wt%		2 - 8	2 - 8	2 - 8	2 - 8	1	Owner	41		B
	Feed Pebble PSD												
	F_{100}		mm		95.0	95.0	95.0	95.0	1	Owner	41	Linear interpolation from PSD data	B
	F_{80}		mm		49.0	49.0	49.0	49.0	1	Owner	41	Linear interpolation from PSD data	B
	F_{50}		mm		37.0	37.0	37.0	37.0	1	Owner	41	Linear interpolation from PSD data	B
	Product Pebble PSD												
	P_{80}		mm		12.0	12.0	12.0	12.0	1	Owner	69	FLS guaranteed value	B
4.6	**Classification**												
	Feed Size F_{50}	Typical	µm		276	276	276	276	1	Owner	16		A
	Feed Size F_{100}	Modelled	mm		350	350	350	350	1	Owner	9	Consultant Modelling	B
	Feed Size F_{80}	Modelled	mm		0.757	0.757	0.757	0.757	1	Owner	9	Consultant Modelling	B
	Feed Size F_{50}	Modelled	mm		0.306	0.306	0.306	0.306	1	Owner	9	Consultant Modelling	B
	Product Size P_{80}	Target	µm		180	180	180	220	1	Owner	26		A
	Product Size P_{80}	Maximum	µm		250	250	250	250	1	Owner	26		A
	Type				Hydrocyclone	Hydrocyclone	Hydrocyclone	Hydrocyclone	1	Owner	16		A
	Cyclone Model				800CVX13	800CVX13	800CVX13	800CVX13	5	Vendor	28		A
	Number of Cyclone Clusters				1	1	1	1	5	Vendor	28		A
	Distributor Outlets per Cluster		No.		18	18	18	18	5	Vendor	28		B
	Number Installed Cyclones		No.		17	17	17	17	5	Vendor	28	15 for primary cyclone duty, 2 for gravity duty	A
	Number Spare Outlets		No.		1	1	1	1	6	Calculated		1 blanked nozzle for the ladder access, 1 spare blanked nozzle for future	
	Number Operating Cyclones - Primary Cluster	Nominal	No.		12	11	12	12	5	Vendor	28		A
		Maximum	No.		14	14	14	14	5	Vendor	28		A
		Standby	No.		1	1	1	1	5	Vendor	28		A
		Total Installed	No.		15	15	15	15	6	Calculated			

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um					
	Number Operating Cyclones - Gravity	Nominal	No.		2	2	2	2	5	Vendor	28	Gravity cyclones require larger spigot and VF to achieve mass flow of 1000tph to underflow. There is also a higher recirc. load.	A
		Maximum	No.		2	2	2	2	5	Vendor	28		A
	Cyclone Feed	Nominal	t/h		14,421	13,986	14,421	14,421	6	Calculated			
	Cyclone Feed Flow	Nominal	m³/h		13,233	12,816	13,217	13,217	6	Calculated			
	Cyclone Feed Flow	Maximum @ 61.6%	m³/h		13,862	13,862	13,862	13,862	1	Owner	26	Flow rate of 13,862m3/hr requires 14 or 15 online, 2 gravity as advised by Weir	A
	Cyclone Feed Density		% w/w		62.0	62.0	62.0	65.0	5	Vendor	28		A
	Underflow Solids Concentration	Minimum	% w/w		75.0	75.0	75.0	75.0	1	Owner	16		B
	Underflow Solids Concentration		% w/w		77.2	77.0	77.2	77.2	6	Calculated			
	Underflow Solids		t/h		11,275	11,024	11,275	11,129	6	Calculated			
	Overflow Solids Concentration		% w/w		36.4	35.9	36.4	43.3	5	Vendor	28		A
	Overflow Solids		t/h		3,146	2,962	3,146	3,494	6	Calculated			
	Circulating Load - Cyclone UF / Cyclone OF		%		358	372	358	318	5	Vendor	28		A
	Circulating Load - Ball Mill Discharge / Cyclone OF		%		301	311	301	267	6	Calculated			
	Operating Pressure - Cyclones in Cluster		kPa		82	91	82	73	5	Vendor	28		B
	Operating Pressure - Gravity Cyclones		kPa		83	86	83	70	5	Vendor	28		B
4.7	**Trash Screen**												
	Type				Linear Screen	Linear Screen	Linear Screen	Linear Screen	1	Owner	16		A
	Total Number Screens				4	4	4	4	6	Calculated			
	Number Operating				3	3	3	3	1	Owner	14		A
	Number Standby				1	1	1	1	1	Owner	14		A
	Screen Dimensions	W x L	m		9.53 x 4.2	9.53 x 4.2	9.53 x 4.2	9.53 x 4.2	5	Vendor	55		B
	Screen Area per Screen	Effective	m²		40.03	40.03	40.03	40.03	5	Vendor	55		A
	Total Screen Area		m²		120.09	120.09	120.09	120.09	6	Calculated			
	Effective Screen Area		%		34	34	34	34	5	Vendor	55		B
	Screen Aperture		mm		1.0 x 1.0	1.0 x 1.0	1.0 x 1.0	1.0 x 1.0	5	Vendor	55		B
	Screen Feed Flow (incl. spray water)		m³/h		2,183	2,093	2,182	1,936	6	Calculated			
	Maximum Screen Flux Rate		m³/m²/h		75	75	75	75	1	Owner	26		A
	Specific Screen Flux Rate		m³/m²/h		54.5	52.3	54.5	48.4	6	Calculated			
	Spray/Wash Water	per screen	m³/h		60	60	60	60	5	Vendor	31, 55		B
	Total Spray/Wash Water		m³/h		180	180	180	180	6	Calculated			
	Spray Water reporting to Screen Oversize	per screen	m³/h		45	45	45	45	5	Vendor	31		B
	Total Spray Water reporting to Screen Oversize	Total			135	135	135	135	6	Calculated			
	Spray Water reporting to Screen Undersize	per screen	m³/h		15	15	15	15	5	Vendor	31		B
	Spray Water reporting to Screen Undersize	Total			45	45	45	45	6	Calculated			
	Water to Trash Screens Final Collection Box		m³/h		258.0	134.9	258.0	306.8	6	Calculated		HP Process Water, back calculated in SysCAD	B
4.8	**Trash Dewatering**												
	Dewatering Screen Type				Vibrating Screen	Vibrating Screen	Vibrating Screen	Vibrating Screen	1	Owner	16		A
	Number				1	1	1	1	1	Owner	16		A
	Screen Size	W x L	m		1.83 x 3.66	1.83 x 3.66	1.83 x 3.66	1.83 x 3.66	5	Vendor	59		B
	Effective Screening Area per Screen		m²		6.14	6.14	6.14	6.14	5	Vendor	59		B
	Specific Screen Flux Rate	Design	m³/m²/h		50	50	50	50	1	Owner	16		A
		Nominal	m³/m²/h		22.36	22.36	22.36	22.36	5	Vendor	59	Includes wash water.	
	Screen Aperture		mm		1 x 15	1 x 15	1 x 15	1 x 15	1	Owner	16		A
	Deck Type				Polyurethane	Polyurethane	Polyurethane	Polyurethane	5	Vendor	59		A
	Quantity of Wood/Grit/Trash removed per day		t/day		27	27	27	27	1	Owner	16		
	Total Spray/Wash Water		m³/h		11	11	11	11	5	Vendor	59		B
5.0	**GRAVITY CONCENTRATION**												
5.1	**Gravity Scalping Screen**												
	Type				Horizontal Vibrating	Horizontal Vibrating	Horizontal Vibrating	Horizontal Vibrating	5	Vendor	59		A
	Number of Screens				2	2	2	2	1	Owner	16	Only 2x gravity concentrator circuits, no future allowance for additional circuits	A
	Screen Solids Feed Rate per Screen	Nominal	t/h		1,000	1,000	1,000	1,000	6	Calculated			
		Design	t/h		1,111	1,111	1,111	1,111	6	Calculated		To achieve 1,000 t/h to gravity concentrators at design O/S split	A
	Screen Feed Density	Target			60.0	60.0	60.0	60.0	1	Owner	70	Excludes Spray Water	B
	Screen Feed Density	Actual	% w/w		60.0	60.0	60.0	60.0	6	Calculated			
	Screen Feed Flowrate per Screen	Nominal	m³/h		968	966	967	967	6	Calculated		Excludes Spray Water	
		Design	m³/h		1,075	1,073	1,074	1,075	6	Calculated		Excludes Spray Water	
	Screen Size	W x L	m		3.66 x 6.10	3.66 x 6.10	3.66 x 6.10	3.66 x 6.10	1	Owner	59, 70, 72		B
	Effective Screen Area per Screen	Overall	m²		22.31	22.31	22.31	22.31	5	Vendor	59		
	Specific Screen Flux Rate	Nominal	m³/m²/h		48.0	47.9	47.9	47.9	6	Calculated		Includes Spray Water	
		Design	m³/m²/h		52.8	52.7	52.7	52.7	6	Calculated		Includes Spray Water	
		Target	m³/m²/h		50	50	50	50	1	Owner	16, 70		A
	Screen Split to Oversize - Range		%		8 - 15	8 - 15	8 - 15	8 - 15	1	Owner	16		A
		Design	%		10	10	10	10	1	Owner	16, 72		A
	Screen Aperture		mm		3.2 x 18mm SWF first half & 2.0 x 18mm SWF second half	3.2 x 18mm SWF first half & 2.0 x 18mm SWF second half	3.2 x 18mm SWF first half & 2.0 x 18mm SWF second half	3.2 x 18mm SWF first half & 2.0 x 18mm SWF second half	5	Vendor	59	3.2mm Slotted first half, 2mm non-pegging second half (VR) requested by Owner in source 16	B
	Screen Spray Water	per screen	m³/h		102	102	102	102	5	Vendor	59		B
	Oversize Solids Concentration		% w/w		87	87	87	87	5	Vendor	59		B
	Oversize Slurry Destination				Ball mill feed	Ball mill feed	Ball mill feed	Ball mill feed	1	Owner	43		A
	Undersize Slurry Destination				Gravity Concentrators	Gravity Concentrators	Gravity Concentrators	Gravity Concentrators	1	Owner	43		A
5.2	**Gravity Concentrator**												
	Type				Knelson	Knelson	Knelson	Knelson	1	Owner	16		A
	Number of Concentrators				2	2	2	2	1	Owner	16	Only 2x gravity concentrator circuits	A
	Model				KC-QS70	KC-QS70	KC-QS70	KC-QS70	1	Owner	16		A
	Feed Capacity per Unit	Max	t/h		1,000	1,000	1,000	1,000	5	Vendor	39, 71		A
	Volumetric Capacity per Unit	Max	m³/h		910	910	910	910	5	Vendor	7		B
	Feed Solids Rate to Circuit		t/h		1,800	1,800	1,800	1,800	2, 5	EPC Vendor	7, 71	300-1000t/hr/unit operating range	A
		Max	t/h		2,000	2,000	2,000	2,000	6	Calculated			
	Feed Solids Rate per Unit	Nominal	t/h		900	900	900	900	6	Calculated		Vendor recommends 90% of the max (Source 71).	
	Feed Density		% w/w		54.2	54.2	54.2	54.2	6	Calculated			
	Volumetric Flowrate per Unit		m³/h		1,021	1,020	1,020	1,021	6	Calculated			
	Fluidising Water per Unit		m³/h		120	120	120	120	5	Vendor	7		B
	Fluidising Water	Nominal	m³/h		240	240	240	240	6	Calculated		120m3 per concentrator	
	Cycle Time (Inclusive of Flush Cycle)		min		30	30	30	30	5	Vendor	71		B
	Cycles Per Day		No.		48	48	48	48	6	Calculated			
	Concentrate Production Per Cycle		kg		150	150	150	150	5	Vendor	7, 71		A
	Concentrate Bulk SG		t/m³		2.20	2.20	2.20	2.20	1	Owner	44		B
	Daily Concentrate Production		kg/d		14,400	14,400	14,400	14,400	6	Calculated			
			m³/d		6.55	6.55	6.55	6.55	6	Calculated			
	Fresh Feed to Concentrators		%		57.2	60.8	57.2	51.5	6	Calculated			
	Tailings Destination				Cyclone Feed Hopper	Cyclone Feed Hopper	Cyclone Feed Hopper	Cyclone Feed Hopper	1	Owner	43		A
6.0	**INTENSIVE LEACH**												
6.1	**Intensive Leach Reactor**												
	Type				Acacia	Acacia	Acacia	Acacia	1	Owner	16		A
	Model				CS10000	CS10000	CS10000	CS10000	1	Owner	70		B
	Number				1	1	1	1	1	Owner	16		A
	Batches per Day	Nominal			1	1	1	1	5	Vendor	39		A
		Max			2	2	2	2	5	Vendor	39	Required to complete 2 batches per day as per Owner Source 72	A
	Solids Feed Rate per Batch		kg/batch		14,400	14,400	14,400	14,400	6	Calculated			
	Design Solids Feed Rate per Batch		kg/batch		15,000	15,000	15,000	15,000	5	Vendor	71		B
	Feed % Solids (after dewatering)		% w/w		90	90	90	90	5	Vendor	71		B
	Gravity Gold Recovery	Nominal	% Au		14	14	14	14	1	Owner	16, 72	Relative to Feed Grade	B
		Design	% Au		20	20	20	20	1	Owner	16, 72	Relative to Feed Grade	A
	Gravity Gold Recovery to Concentrate per Day	Nominal	g Au		16,701	18,015	18,015	18,550	6	Calculated			
		Design	g Au		23,858	25,736	25,736	26,500	6	Calculated			
	Gold Feed Grade	Nominal	ppm Au		1,160	1,251	1,251	1,288	6	Calculated			
		Design	ppm Au		1,591	1,716	1,716	1,767	6	Calculated			
	Gravity Leach Recovery	Range	%		92 - 98	92 - 98	92 - 98	92 - 98	1	Owner	16		A
		Design	%		98	98	98	98	1	Owner	16, 72	Value used in SysCAD model	B
	Cycle Time		h		12	12	12	12	1	Owner	16		A
	Pregnant Solution Volume	per batch	m³		14.5	14.5	14.5	14.5	5	Vendor	71		B
	Pregnant Solution Grade	Nominal	ppm Au		1,129	1,218	1,218	1,254	6	Calculated			
		Design	ppm Au		1,613	1,739	1,739	1,791	6	Calculated			
		Average	ppm Ag		29.3	29.3	29.3	29.3	1	Owner	33	Plant assays from Client	B
		Average	ppm Cu		302.9	302.9	302.9	302.9	1	Owner	33	Plant assays from Client	B
	Scheme Water Required	per batch	m³		12	12	12	12	6	Calculated			
	Residue (Tailings) % Solids		%w/w		40.0	40.0	40.0	40.0	5	Vendor	71		B
6.2	**Gravity Electrowinning**												
	Type				Clean in place	Clean in place	Clean in place	Clean in place	1	Owner	16		A
	Number of Circuits		No.		1	1	1	1	1	Owner	16		A
	Number Cells per Circuit		No.		1	1	1	1	1	Owner	16		A
	Number of Acacia Batches per day		No.		2	2	2	2	1	Owner	70		
	Electrowinning Time		h		10.0	10.0	10.0	10.0	5	Vendor	33, 72, 73	Two electrowinning runs as required per day as per Owner Source 70 & 72	A
	Barren Solution Assay		ppm Au		<5	<5	<5	<5	1	Owner	16, 70		A
	Electrowinning Cell	Model			1000x1000-22D split	1000x1000-22D split	1000x1000-22D split	1000x1000-22D split	5	Vendor	73	2 compartments	B
	Number of Cathodes				22	22	22	22	5	Vendor	73	11 per compartment	B
	Feed Flowrate		m³/h		48	48	48	48	5	Vendor	73	24m3/h to each compartment for a 48m3/h total. Vendor advised this is dependent on silver and copper grades. This may need to be reduced for higher silver grades to reduce detachment.	B
	Cell Linear Velocity		m/min		0.4	0.4	0.4	0.4	5	Vendor	73	Vendor advised this is dependent on silver and copper grades	B
	Current Density		A/m²		15-25	15-25	15-25	15-25	5	Vendor	73		B
	Rectifier	per compartment	A		2,500	2,500	2,500	2,500	5	Vendor	73	2 off, one per compartment	B

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um					
7.0	**FLOTATION**												
7.1	**General**												
	Throughput		t/h		3,146	2,962	3,146	3,494	6	Calculated			
	Feed Particle Size - p80		µm		180	180	180	220	1	Owner	26		A
		Design	µm		220	220	220	220	1	Owner	26		B
	Froth Factors												
	Rougher Flotation Concentrate				2.00	2.00	2.00	2.00	1	Owner	16		A
	Scavenger Flotation Concentrate				1.60	1.60	1.60	1.60	1	Owner	16		A
	Cleaner Flotation Concentrate				2.50	2.50	2.50	2.50	1	Owner	16		A
	Cleaner Scavenger Flotation Concentrate				2.00	2.00	2.00	2.00	1	Owner	16		A
	Flotation Cell Transfer Lines				1.20	1.20	1.20	1.20	1	Owner	16		A
	Launders				-	-	-	-	1	Owner	16	Open launders - design to ensure no overflow	A
	Spray Water Line Pressure (at vendor battery limit)		kPa		82.74	82.74	82.74	82.74	5	Vendor	60		B
7.2	**Rougher Flotation**											Owner Supplied	
	Feed Streams				Trash Screen Underflow	Trash Screen Underflow	Trash Screen Underflow	Trash Screen Underflow	1	Owner	43		A
	Solids Feed Rate	New Feed	t/h		3,146	2,962	3,146	3,494	6	Calculated			
		Total	t/h		3,161	2,977	3,160	3,512	6	Calculated		Includes sample return	
	Volumetric Feed Rate		m³/h		6,719	6,326	6,716	6,027	6	Calculated			
	Feed Solids Concentration		% w/w		34.4	34.4	34.4	40.5	6	Calculated			
	Particle Size p80	Design	µm		220	220	220	220	2	EPC	63	Feed, concentrate and tailings p80 assumed equal (conservative)	B
	Mass Pull		%		4.4	4.4	4.4	4.4	6	Calculated			
	Concentrate 1 Upgrade Ratio				15	15	15	15	1	Owner	16		A
	Concentrate 1 Grade		% S		15.3	22.4	21.0	15.3	6	Calculated			
	Concentrate 1 Stage Recovery		% Au in fl. feed		65	65	65	65	1	Owner	23		A
	Concentrate 1 Stage Recovery		% of S in fl. feed		65	65	65	65	1	Owner	23		A
	Concentrate 1 Production		t/h		138	130	138	153	6	Calculated			
	Concentrate 1 Solids Concentration		% w/w		25	25	25	25	1	Owner	16		A
	Concentrate 1 Slurry Volume - Excl Spray Water		m³/h		427	400	425	474	6	Calculated			
	Concentrate 1 SG				3.32	3.54	3.50	3.32	2	EPC	32		A
	Concentrate 1 Launder Spray Water	Operating Range	m³/h		6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	5	Vendor	60		B
		Design	m³/h		50	50	50	50	5	Vendor	60		A
	Mass Pull		%		2.4	2.4	2.4	2.4	6	Calculated			
	Concentrate 2 & 3 Upgrade Ratio				7	7	7	7	1	Owner	16		A
	Concentrate 2 & 3 Grade		% S		7.1	10.4	9.8	7.1	6	Calculated			
	Concentrate 2 & 3 Stage Recovery		% Au in fl. feed		17	17	17	17	1	Owner	23		A
	Concentrate 2 & 3 Stage Recovery		% of S in fl. feed		17	17	17	17	1	Owner	23		A
	Concentrate 2 & 3 Production		t/h		77	72	77	86	6	Calculated			
	Concentrate 2 & 3 Solids Concentration		% w/w		25	25	25	25	1	Owner	16		A
	Concentrate 2 & 3 Slurry Volume - Excl Spray Water		m³/h		241	225	239	268	6	Calculated			
	Concentrate 2 & 3 SG				3.06	3.17	3.15	3.06	2	EPC	32		A
	Concentrate 2 Launder Spray Water	Operating Range	m³/h		6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	5	Vendor	60		
	Concentrate 3 Launder Spray Water	Operating Range	m³/h		6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	5	Vendor	60		
	Concentrate 2 & 3 Spray Water	Design (Combined)	m³/h		50	50	50	50	5	Vendor	50		A
	Rougher Tail Solids Rate		t/h		2,946	2,775	2,946	3,273	6	Calculated			
	Rougher Tail Slurry Volumetric Flow		m³/h		6,051	5,701	6,052	5,285	6	Calculated			
	Cell Type				Tank Cells	Tank Cells	Tank Cells	Tank Cells	1	Owner	16		A
	Aspiration				Forced	Forced	Forced	Forced	1	Owner	16		A
	Number Cells Installed				3	3	3	3	1	Owner	41		A
	Nominal Cell Volume		m³		630	630	630	630	1	Owner	16		A
	Total Cell Volume		m³		1,890	1,890	1,890	1,890	6	Calculated			
	Cell Air Hold-Up		%		15	15	15	15	1	Owner	41		A
	Plant Residence Time		min		15.1	16.0	15.1	17.0	6	Calculated		Rougher residence time existing plant = 10-12 mins (source 23). Residence time has been calculated using the average of the feed and tailings flowrates.	
	Lip Loading - Design		t/m/h		1.5	1.5	1.5	1.5	1	Owner	16		A
	Froth Carry Rate - Design		t/m²/h		1.5	1.5	1.5	1.5	1	Owner	16		A
7.3	**Scavenger Flotation**											Owner Supplied	
	Feed Streams				Rougher Tailings	Rougher Tailings	Rougher Tailings	Rougher Tailings	1	Owner	43		A
	Solids Feed Rate		t/h		2,946	2,775	2,946	3,273	6	Calculated			
	Volumetric Feed Rate		m³/h		6,051	5,701	6,052	5,286	6	Calculated			
	Feed Solids Concentration		% w/w		35.3	35.3	35.3	42.4	6	Calculated			
	Particle Size p80	Design	µm		220	220	220	220	2	EPC	63	Feed, concentrate and tailings p80 assumed equal (conservative)	
	Mass Pull (Scavenger Concentrate)		%		1.1	1.1	1.1	1.1	6	Calculated			
	Scavenger Concentrate Upgrade Ratio				5	5	5	5	1	Owner	16		A
	Scavenger Concentrate Grade		% S		5.1	7.5	7.0	5.1	6	Calculated			
	Scavenger Concentrate Stage Recovery		% Au in fl. feed		5	5	5	5	1	Owner	23		A
	Scavenger Concentrate Stage Recovery		% of S in fl. feed		5	5	5	5	1	Owner	23		A
	Scavenger Concentrate Production		t/h		32	30	32	35	6	Calculated			
	Scavenger Concentrate Solids Concentration		% w/w		25	25	25	25	2	EPC	32		A
	Scavenger Concentrate Slurry Volume - Excl Spray Water		m³/h		99	93	99	110	6	Calculated			
	Scavenger Concentrate SG				3.00	3.07	3.06	3.00	2	EPC	32		A
	Scavenger Concentrate Solids Concentration - Diluted		% w/w		20	20	20	20	2	EPC	32		A
	Scavenger Concentrate Launder Spray Water	Actual (Combined)	m³/h		29	28	29	33	6	Calculated			
		Operating Range	m³/h/cell		6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	5	Vendor	60		B
		Design	m³/h/cell		25.00	25.00	25.00	25.00	5	Vendor	60		B
	Scavenger Tail Solids Rate		t/h		2,914	2,745	2,914	3,238	6	Calculated			
	Scavenger Tail Slurry Volumetric Flow		m³/h		5,952	5,608	5,953	5,176	6	Calculated			
	Cell Type				Tank Cells	Tank Cells	Tank Cells	Tank Cells	1	Owner	16		A
	Aspiration				Forced	Forced	Forced	Forced	1	Owner	16		A
	Number Cells Installed				4	4	4	4	1	Owner	41		A
	Nominal Cell Volume		m³		630	630	630	630	1	Owner	16		A
	Total Cell Volume		m³		2,520	2,520	2,520	2,520	6	Calculated			
	Cell Air Hold-Up		%		15	15	15	15	1	Owner	41		A
	Plant Residence Time		min		21.4	22.7	21.4	24.6	6	Calculated		Scavenger residence time existing plant = 29 - 32 mins (source 23). Residence time has been calculated using the average of the feed and tailings flowrates.	
	Lip Loading - Design		t/m/h		1.5	1.5	1.5	1.5	1	Owner	16		A
	Froth Carry Rate - Design		t/m²/h		1.5	1.5	1.5	1.5	1	Owner	16		A
	Combined Rougher and Scavenger Residence Time				36.5	38.8	36.5	41.6	6	Calculated			
7.4	**Cleaner Flotation**											Owner Supplied	
	Feed Streams				Rougher and Scavenger Concentrate Cleaner Scavenger Concentrate	Rougher and Scavenger Concentrate Cleaner Scavenger Concentrate	Rougher and Scavenger Concentrate Cleaner Scavenger Concentrate	Rougher and Scavenger Concentrate Cleaner Scavenger Concentrate	1	Owner	43		A
	Solids Feed Rate	New Feed Only	t/h		253	244	258	282	6	Calculated		New feed only	
		Total	t/h		465	424	439	478	6	Calculated		Total feed to Jameson including tails recycle and sample return minus cleaner feed sample to OSA	
	Volumetric Feed Rate to Cell - Target	Total	m³/h		1,998	1,998	1,998	1,998	5	Vendor	27	Average of four modelled cases	C
	Volumetric Feed Rate to Cell - Actual	Total	m³/h		1,998	1,998	1,998	1,998	6	Calculated		Total feed to Jameson including tails recycle and sample return minus cleaner feed sample to OSA	
	Feed Solids Concentration - Actual	Total	% w/w		19.0	17.5	18.0	19.5	6	Calculated		Total feed to Jameson including tails recycle and sample return minus cleaner feed sample to OSA	
	Particle Size p80	Design	µm		220	220	220	220	6	Calculated		Feed, concentrate and tailings p80 assumed equal (conservative). Max of calculated p80	
		Calculated	µm		180	180	180	220	6	Calculated			
	Mass Pull		%		30.8	44.0	41.5	30.7	6	Calculated			
	Concentrate Grade		% S		35	35	35	35	1	Owner	26		A
	Concentrate Stage Recovery		% S		93.3	93.6	93.4	93.1	6	Calculated			
	Concentrate Recovery	Overall	% of S in fl. feed		85.1	85.1	85.1	85.1	1	Owner	23		A
	Concentrate Recovery	Overall	% Au in fl. feed		84	83	83	84	6	Calculated			
	Concentrate Production		t/h		78	107	107	86	6	Calculated			
	Concentrate Solids Concentration		% w/w		25	25	25	25	5	Vendor	27	Undiluted cleaner flotation concentrate	A
	Concentrate Slurry Volume - Excl Spray Water		m³/h		237	326	326	263	6	Calculated			
	Concentrate SG				3.93	3.93	3.93	3.93	2	EPC	32		A
	Concentrate Wash Water Ratio (of Con m3/hr)				1.0	1.0	1.0	1.0	5	Vendor	27		A
	Cleaner Tail Solids Rate		t/h		183	145	159	203	6	Calculated		To Cleaner-Scavenger Flotation Cell Feed Hopper	
	Cleaner Tail Slurry Volumetric Flow		m³/h		945	910	957	1,038	6	Calculated		To Cleaner-Scavenger Flotation Cell Feed Hopper	
	Cleaner Tails Recycle		%		52.7	54.4	52.1	48.1	6	Calculated		The % of total tailings recycled back to feed hopper.	
	Cleaner Tail Solids Rate		t/h		204	173	173	188	6	Calculated		Recycle back to Cleaner Flotation Cell Feed Hopper	
	Cleaner Tail Slurry Volumetric Flow		m³/h		1,053	1,088	1,041	960	6	Calculated		Recycle back to Cleaner Flotation Cell Feed Hopper	
	Cell Type				Jameson	Jameson	Jameson	Jameson	5	Vendor	18		A
	Model				B6500/24	B6500/24	B6500/24	B6500/24	5	Vendor	27		A
	Aspiration Type				Natural	Natural	Natural	Natural	5	Vendor	16		A
	Number Cells Installed				1	1	1	1	5	Vendor	18		A
	Number of Downcomers				24	24	24	24	5	Vendor	27		B
	Froth Area		m²		22.2	22.2	22.2	22.2	5	Vendor	27		A
	Froth Carry Rate	Nom	t/m²/h		3.50	4.81	3.50	3.51	5	Vendor	27		B
		Max	t/m²/h		6.5	6.5	6.5	6.5	5	Vendor	27		A
		Calculated	t/m²/h		3.51	4.82	4.82	3.89	6	Calculated			

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um					
7.5	**Cleaner Scavenger Flotation**												
	Feed Streams				Cleaner Flotation Tail	Cleaner Flotation Tail	Cleaner Flotation Tail	Cleaner Flotation Tail	1	Owner	66	Owner Supplied	B
	Solids Feed Rate	New Feed Only	t/h		183	145	159	203	6	Calculated		New feed only	
		Total	t/h		380	300	317	385				Total feed to Jameson cell including tails recycle	
	Volumetric Feed Rate to Cell - Target	Total	m^3/h		2,048	2,048	2,048	2,048	5	Vendor	27	Average of four modelled cases	C
	Volumetric Feed Rate - Actual	Total	m^3/h		2,048	2,048	2,048	2,048	6	Calculated		Total feed to Jameson cell including tails recycle	
	Feed Solids Concentration	Total	% w/w		15.6	12.6	13.2	15.8	6	Calculated		Total feed to Jameson cell including tails recycle	
	Particle Size p80	Design	µm		220	220	220	220	2	EPC	63	Feed, concentrate and tailings p80 assumed equal (conservative). Max of calculated p80	
	Mass Pull		%		7.6	13.3	12.1	7.6	6	Calculated			
	Concentrate Grade		% S		13.10	13.10	13.10	13.10	5	Vendor	19	Design Grade sulphur grade based on EPC circuit modelling	A
	Concentrate Stage Recovery		% S		49.6	54.1	54.2	50.9	6	Calculated			A
	Concentrate Recovery		% of Au in fl. feed		5.7	5.7	5.7	5.7	3	Assumed		Assumed to match the S recovery as seen in other areas of the flotation circuit	A
	Concentrate Recovery		% of S in fl. feed		5.7	5.7	5.7	5.7	2	EPC	32		A
	Concentrate Production		t/h		14	19	19	15	6	Calculated			
	Concentrate Solids Concentration		% w/w		25	25	25	25	2	EPC	32		B
	Concentrate Slurry Volume - Excl Spray Water		m^3/h		43	60	60	48	6	Calculated			
	Concentrate SG				3.25	3.25	3.25	3.25	2	EPC	32		A
	Concentrate Wash Water Ratio (of Con m3/hr)				1	1	1	1	5	Vendor	27		A
	Cleaner Scavenger Tail Solids Rate		t/h		169	125	140	188	6	Calculated		To Cleaner-Scavenger Flotation Cell Tails Hopper	
	Cleaner Scavenger Tail Slurry Volumetric Flow		m^3/h		948	914	960	1,041	6	Calculated		To Cleaner-Scavenger Flotation Cell Tails Hopper	
	Cleaner Scavenger Tails Recycle		%		53.7	55.4	53.1	49.2	6	Calculated		The % of total tailings recycled back to feed hopper.	
	Cleaner Scavenger Tail Solids Rate		t/h		196	156	158	182	6	Calculated		Recycle back to Cleaner Flotation Cell Feed Hopper	
	Cleaner Scavenger Tail Slurry Volumetric Flow		m^3/h		1,100	1,134	1,088	1,007	6	Calculated		Recycle back to Cleaner Flotation Cell Feed Hopper	
	Cell Type				Jameson	Jameson	Jameson	Jameson	5	Vendor	19		A
	Model				B6500/24	B6500/24	B6500/24	B6500/24	1	Owner	27		A
	Aspiration Type				Natural	Natural	Natural	Natural	1	Owner	16		A
	Number Cells Installed				1	1	1	1	5	Vendor	19		A
	Number of Downcomers				24	24	24	24	5	Vendor	27		B
	Froth Area		m^2		22.2	22.2	22.2	22.2	5	Vendor	27		A
	Froth Carry Rate	Nom	$t/m^2/h$		0.63	0.85	0.63	0.63	5	Vendor	27		B
		Max	$t/m^2/h$		6.5	6.5	6.5	6.5	5	Vendor	27		A
		Calculated	$t/m^2/h$		0.63	0.86	0.86	0.70	6	Calculated			
8.0	**CONCENTRATE HANDLING**												
8.1	**Concentrate Thickener (Brownfields)**				Existing Flotation Tail Pre-Leach Thickener (CIL3)	Existing Flotation Tail Pre-Leach Thickener (CIL3)	Existing Flotation Tail Pre-Leach Thickener (CIL3)	Existing Flotation Tail Pre-Leach Thickener (CIL3)				Existing 30-TH-61	
	Feed Stream				Final Concentrate Hopper (Cleaner Flotation Concentrate)	Final Concentrate Hopper (Cleaner Flotation Concentrate)	Final Concentrate Hopper (Cleaner Flotation Concentrate)	Final Concentrate Hopper (Cleaner Flotation Concentrate)	1	Owner	16		A
	Solids Feed Rate		t/h		75	104	104	84	6	Calculated			
	Feed Flowrate - Excl. Diluted Flocculant		m^3/h		230	319	319	256	6	Calculated			
	Feed Solids Concentration		% w/w		24.9	24.9	24.9	24.9	6	Calculated			
	Underflow Solids Concentration		% w/w		60	60	60	60	1	Owner	16		A
	Testwork Flux Rate		$t/m^2/h$		0.25	0.25	0.25	0.25	1	Owner	23		A
	Calculated Flux Rate		$t/m^2/h$		0.13	0.18	0.18	0.15	6	Calculated			
	Thickener Area Calculated		m^2		573	573	573	573	6	Calculated			
	Diameter	Existing	m		27	27	27	27	1	Owner	16		A
	Structural Design SG				3.8	3.8	3.8	3.8	1	Owner	16		A
	Concentrate Thickener Sprays - Mill Circuit Water		m^3/h		68.40	68.40	68.40	68.40	5	Vendor	113		1
8.2	**Filtered Feed Tanks A & B (Brownfields)**				Existing	Existing	Existing	Existing					
	Feed Streams				Concentrate Thickener Underflow	Concentrate Thickener Underflow	Concentrate Thickener Underflow	Concentrate Thickener Underflow	1	Owner	43	Shared volume between Filtered Feed Tank A & B via balance line.	A
	Solids Feed Rate		t/h		75	104	104	84	6	Calculated			
	Volumetric Feed Rate		m^3/h		66	92	92	74	6	Calculated			
	Feed Solids Concentration		% w/w		59.6	59.7	59.7	59.6	6	Calculated			
	Number of Tanks				2	2	2	2	1	Owner	16		A
	Residence Time		h		21.2	15.3	15.3	19.0	6	Calculated			
	Tank Capacity	per tank	m^3		700	700	700	700	1	Owner	16		A
8.3	**Concentrate Filters (Brownfields)**				Existing	Existing	Existing	Existing				Remains in operation to treat additional concentrate in excess of the UFG capacity.	
	Feed Stream				Concentrate Filtered Feed Tank B	Concentrate Filtered Feed Tank B	Concentrate Filtered Feed Tank B	Concentrate Filtered Feed Tank B	1	Owner	43		A
	Filter Type				Vacuum Belt Filter	Vacuum Belt Filter	Vacuum Belt Filter	Vacuum Belt Filter	1	Owner	43		A
	Number of Filters				2	2	2	2	1	Owner	16		A
	Solids Feed Rate		t/h		0	0	0	0	6	Calculated		Solids have been prioritised through UFG circuits first.	
	Volumetric Feed Rate		m^3/h		0	0	0	0	6	Calculated			
	Feed Solids Concentration	Target	% w/w		60.0	60.0	60.0	60.0	1	Owner	16		A
		Actual	% w/w		0.0	0.0	0.0	0.0	6	Calculated			
	Target Moisture Content		<% w/w		10.0	10.0	10.0	10.0	1	Owner	16		A
	Filter Area	per filter	m^2		55.0	55.0	55.0	55.0	1	Owner	16		A
	Specific Filtration Rate	Nominal	$m^3/m^2/h$		0.14	0.14	0.14	0.14	1	Owner	16		A
		Design	$m^3/m^2/h$		0.31	0.31	0.31	0.31	1	Owner	16		A
8.4	**Concentrate Stockpile**											Remains in operation to treat additional concentrate in excess of the UFG capacity.	
	Feed Stream				Concentrate Filter Cake	Concentrate Filter Cake	Concentrate Filter Cake	Concentrate Filter Cake	1	Owner	43		A
	No. of Stockpiles				1	1	1	1	1	Owner	16		A
	Concentrate Bulk Density		t/m^3		2.2	2.2	2.2	2.2	1	Owner	16		A
	Angle of Repose	Nominal	Degrees		40.0	40.0	40.0	40.0	1	Owner	16		A
	Storage		days		2	2	2	2	1	Owner	16		A
	Capacity		m^3		1,139	1,139	1,139	1,139	1	Owner	16		A
	TML for Bulk Shipment		% w/w		6.10 - 8.26	6.10 - 8.26	6.10 - 8.26	6.10 - 8.26	1	Owner	23		A
9.0	**CONCENTRATE ULTRAFINE GRINDING**												
9.1	**Concentrate UFG Mill 1 (Brownfields)**				Existing	Existing	Existing	Existing				M3000	
	Circuit Feed Streams				Concentrate Filter Feed Tank A	Concentrate Filter Feed Tank A	Concentrate Filter Feed Tank A	Concentrate Filter Feed Tank A	1	Owner	16, 80		1
	Feed Rate - Fresh Feed	P_{80} 9 µm	t/h		0.0	10.0	10.0	0.0	1	Owner	69		B
		Design	t/h		0.0	13.0	13.0	0.0	1	Owner	16		A
	Feed Solids Concentration		% w/w		45	45	45	45	1	Owner	4		A
	Feed F_{80}	Nominal	µm		144	144	144	144	1	Owner	16		A
	Specific Energy	P_{80} 9 µm	kWh/t		116.5	116.5	116.5	116.5	1	Owner	16		A
	Type				IsaMill	IsaMill	IsaMill	IsaMill	1	Owner	16		A
	Model				M3000	M3000	M3000	M3000	1	Owner	16		A
	Number				1	1	1	1	1	Owner	16		A
	Configuration				Closed Circuit	Closed Circuit	Closed Circuit	Closed Circuit	1	Owner	16		A
	Mill Power Draw	Nominal	kW		860	860	860	860	1	Owner	16		A
	Installed Motor Power		kW		1,120	1,120	1,120	1,120	1	Owner	16		A
	UFG Mill Recycle Proportion		%		0	0	0	0	1	Owner	4		A
	UFG Circuit Product Size P_{80}	Nominal	µm		9.0	9.0	9.0	9.0	1	Owner	16		A
	UFG Circuit Product Size P_{80} - Range		µm		9 - 12	9 - 12	9 - 12	9 - 12	1	Owner	16		A
	Gland Seal Water		m^3/h		0.0	1.2	1.2	0.0	1	Owner	2		A
	Flushing Gland Seal Water		m^3/h		0.0	5.4	5.4	0.0	1	Owner	2		A
9.2	**UFG Mill 1 Trash Screen (Brownfields)**				Existing	Existing	Existing	Existing					
	Feed Stream				UFG Mill 1 Discharge	UFG Mill 1 Discharge	UFG Mill 1 Discharge	UFG Mill 1 Discharge	1	Owner	43		A
	Type				Vibrating Screen	Vibrating Screen	Vibrating Screen	Vibrating Screen	1	Owner	16		A
	Number of Screens per UFG Circuit		No.		1	1	1	1	1	Owner	16		A
	Screen Solids Feed Rate per Screen	P_{80} 9 µm	t/h		0	61	61	0	6	Calculated			
	Screen Feed Density	P_{80} 9 µm	% w/w		45.5	42.8	42.8	43.5	6	Calculated			
	Screen Feed Flowrate per Screen	P_{80} 9 µm	m^3/h		0	91	91	0	6	Calculated			
	Screen Size	W x L	m		1.2 x 2.4	1.2 x 2.4	1.2 x 2.4	1.2 x 2.4	1	Owner	16		A
	Screen Area per Screen		m^2		2.9	2.9	2.9	2.9	6	Calculated			
	Specific Screen Flux Rate		$m^3/m^2/h$		0.0	31.6	31.6	0.0	6	Calculated			
	Screen Aperture		mm		1.0	1.0	1.0	1.0	1	Owner	16		A
	Screen Material				Wire Mesh	Wire Mesh	Wire Mesh	Wire Mesh	1	Owner	23		A
	Screen Spray Water		m^3/h		0	0	0	0	1	Owner	44		B
9.3	**UFG Mill 1 Classification (Brownfields)**				Existing	Existing	Existing	Existing					
	Cyclone Model				100CVX	100CVX	100CVX	100CVX	1	Owner	16		A
	Number of Cyclone Clusters per UFG Circuit				1	1	1	1	1	Owner	16		A
	Number Installed Cyclones per Cluster		No.		30	30	30	30	1	Owner	16		A
	Cyclone Solids Feed Rate	P_{80} 9 µm	t/h		0	73	73	0	6	Calculated			
	Cyclone Feed Flow - per UFG circuit	P_{80} 9 µm	m^3/h		0	120	120	0	6	Calculated			
	Cyclone Feed Solids Concentration		% w/w		40	40	40	40	1	Owner	4		A
	Underflow Solids Concentration		% w/w		60	60	60	60	1	Owner	5		A
	Underflow Solids - per UFG circuit	P_{80} 9 µm	t/h		0	61	61	0	6	Calculated			
	Overflow Solids Concentration	P_{80} 9 µm	% w/w		1.1	15.0	15.0	1.1	6	Calculated			
	Overflow Solids - per UFG circuit	P_{80} 9 µm	t/h		0	12	12	0	6	Calculated			
	Cyclone Circulating Load		%		500	500	500	500	1	Owner	16		A
	Operating Pressure		kPa		300	300	300	300	1	Owner	16		A
	Product Size P_{80}		µm		9	9	9	9	1	Owner	5		A

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um					
9.4	**New Concentrate UFG Mill 2 & 3**				Greenfields	Greenfields	Greenfields	Greenfields				M15,000, 2 off	A
	Circuit Feed Streams	UFG Mill 2/3			Concentrate Filter Feed Tank A & B (common balance line)	Concentrate Filter Feed Tank A & B (common balance line)	Concentrate Filter Feed Tank A & B (common balance line)	Concentrate Filter Feed Tank A & B (common balance line)	2	EPC			0
	Nominal Feed Rate - Fresh Feed	P_{80} 9 µm	t/h		30.0	30.0	30.0	30.0	1	Owner	24		A
	Maximum Feed Rate - Fresh Feed	P_{80} 12 µm	t/h		46.0	46.0	46.0	46.0	1	Owner	24		A
	Total Feed Solids Concentration		% w/w		47.5	47.5	47.5	47.7	5	Vendor	65	Includes Recycle Portion	B
	Feed F_{80}	Nominal	µm		144	144	144	144	1	Owner	16		A
	Specific Energy	P_{80} 9 µm	kWh/t		116.5	116.5	116.5	116.5	1	Owner	16		A
	Type				IsaMill	IsaMill	IsaMill	IsaMill	1	Owner	23		A
	Model				M15,000	M15,000	M15,000	M15,000	1	Owner	23		A
	Number				2	2	2	2	1	Owner	23, 43		A
	Configuration				Closed Circuit	Closed Circuit	Closed Circuit	Closed Circuit	1	Owner	23		A
	Mill Power Draw	Nominal	kW		3,500	3,500	3,500	3,500	1	Owner	5		A
	Installed Motor Power		kW		3,800	3,800	3,800	3,800	1	Owner	23		A
	UFG Mill Recycle Proportion		%		20	20	20	5	5	Vendor	65	% of New (Fresh) UFG Mill 2 Feed	B
	UFG Circuit Product Size P_{80}	Nominal	µm		9.0	9.0	9.0	9.0	1	Owner	16		A
	UFG Circuit Product Size P_{80} - Range		µm		9 - 12	9 - 12	9 - 12	9 - 12	1	Owner	16		A
	Gland Seal Water		m^3/h		2.0	2.0	2.0	2.0	5	Vendor	65	Scheme Water	B
	Flushing Gland Seal Water		m^3/h		5.4	5.4	5.4	5.4	5	Vendor	65	Scheme Water	B
9.5	**UFG Mill 2 & 3 Trash Screens**				Greenfields	Greenfields	Greenfields	Greenfields					
	Feed Stream				UFG Mill 2 & 3 Fresh Feed	UFG Mill 2 & 3 Fresh Feed	UFG Mill 2 & 3 Fresh Feed	UFG Mill 2 & 3 Fresh Feed	1	Owner	23		A
	Type				Inclined Static Screen	Inclined Static Screen	Inclined Static Screen	Inclined Static Screen	1	Owner	74		B
	Number of Screens per UFG Circuit		No.		1	1	1	1	1	Owner	23		A
	Screen Solids Feed Rate per Screen	P_{80} 9 µm	t/h		37	46	46	42	6	Calculated			
	Screen Feed Density	P_{80} 9 µm	% w/w		59.6	59.0	59.0	59.6	6	Calculated			
	Screen Feed Flowrate per Screen	P_{80} 9 µm	m^3/h		33	41	41	37	6	Calculated			
	Screen Size	L x W	m		1.240 x 1.6	1.240 x 1.6	1.240 x 1.6	1.240 x 1.6	5	Vendor	77		B
	Screen Area per Screen		m^2		2.84	2.84	2.84	2.84	5	Vendor	77		B
	Specific Screen Flux Rate		$m^3/m^2/h$		14.8	17.7	17.7	16.1	6	Calculated		Includes wash water	
	Screen Aperture		mm		1.0	1.0	1.0	1.0	1	Owner	16		A
	Screen Material				Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	5	Vendor	77		B
	Screen Spray Water		m^3/h		9.0	9.0	9.0	9.0	5	Vendor	77		B
9.6	**UFG Mill 2 & 3 Classification**				Greenfields	Greenfields	Greenfields	Greenfields					
	Cyclone Model				100CVX	100CVX	100CVX	100CVX	1	Owner	16		A
	Number of Cyclone Clusters per UFG Circuit				1	1	1	1	5	Vendor	54		A
	Number Installed Cyclones per Cluster		No.		24	24	24	24	5	Vendor	54		B
	Number of Cyclone Distributor Outlets Available		No.		24	24	24	24	5	Vendor	54		A
	Spare Outlets Available		No.		0	0	0	0	6	Calculated			
	Number of Operating Cyclones		No.		14	14	14	19	5	Vendor	54		B
	Number of Standby Cyclones		No.		10	10	10	5	6	Calculated			
	Cyclone Solids Feed Rate - per UFG circuit	P_{80} 9 µm	t/h		147.3	161.0	161.0	146.1	6	Calculated			
	Cyclone Feed Flow - per UFG circuit	P_{80} 9 µm	m^3/h		319.0	348.5	348.5	315.1	6	Calculated			
	Cyclone Feed Solids Concentration		% w/w		32.8	32.8	32.8	32.9	5	Vendor	65		B
	Underflow Solids Concentration		% w/w		48.0	48.0	48.0	48.0	5	Vendor	65		B
	Underflow Solids - per UFG circuit	P_{80} 9 µm	t/h		109.9	115.0	115.0	104.4	6	Calculated			
	Overflow Solids Concentration	P_{80} 9 µm	% w/w		17.0	18.3	18.3	18.4	6	Calculated			
	Overflow Solids - per UFG circuit	P_{80} 9 µm	t/h		37.5	46.0	46.0	41.8	6	Calculated			
	Cyclone Circulating Load		%		293	250	250	250	5	Vendor	54		0
	Operating Pressure		kPa		247	247	247	248	5	Vendor	54		B
	Product Size P_{80}		µm		9	9	9	9	1	Owner	5		A
9.7	**Concentrate Pre-Leach Thickener (Brownfields)**				Existing	Existing	Existing	Existing				Existing Pre-Leach Thickener 30-TH-31 Repurposed	
	Feed Streams				UFG Circuit 1, 2 & 3 Cyclone O/F's	UFG Circuit 1, 2 & 3 Cyclone O/F's	UFG Circuit 1, 2 & 3 Cyclone O/F's	UFG Circuit 1, 2 & 3 Cyclone O/F's	1	Owner	24		A
	Solids Feed Rate		t/h		75	104	104	84	6	Calculated			
	Feed Flowrate - Excl. Diluted Flocculant		m^3/h		362	475	475	368	6	Calculated			
	Feed Solids Concentration		% w/w		17	18	18	18	6	Calculated			
	Underflow Solids Concentration		% w/w		55	55	55	55	1	Owner	115	Operating range of 52 - 58 %w/w	1
	Calculated Flux Rate		$t/m^2/h$		0.13	0.18	0.18	0.15	6	Calculated			
	Testwork Flux Rate		$t/m^2/h$		0.25	0.25	0.25	0.25	1	Owner	103		A
	Thickener Area Calculated		m^2		573	573	573	573	6	Calculated			
	Diameter	Existing	m		27	27	27	27	1	Owner	103		A
	Thickener Spray Water		m^3/h		15	15	15	15	3	Assumed			C
10.0	**CIL - FLOTATION CONCENTRATE**												
10.1	**General Circuit (Brownfields)**				Existing CIL 2 & 3	Existing CIL 2 & 3	Existing CIL 2 & 3	Existing CIL 2 & 3					
	Solids Feed Rate	Fresh	t/h		75	104	104.1	84	6	Calculated		Based on P80 9 µm	
	Feed Pulp Density	Range	% w/w		50 - 60	50 - 60	50 - 60	50 - 60	1	Owner	108		0
	Actual Feed Pulp Density	Total	% w/w		49	49	48.7	49	6	Calculated		Inclusive of reagent	
	Slurry Feed Flow Rate	Fresh	m^3/h		77	107	107	86	6	Calculated		Prior to reagent addition	
	Slurry Feed Flow Rate	Total	m^3/h		99	137	137	110	6	Calculated		Inclusive of reagent	
	Leach Gold Extraction		%		90	90	90	90	9	Agreed		Increased to 90% from 84% (Source 44) as per meeting with Owner & EPC 22/01/25	2
	Adsorption Efficiency		%		99	99	99	99	1	Owner	44		B
10.2	**Leaching (Brownfields)**				Existing CIL 2	Existing CIL 2	Existing CIL 2	Existing CIL 2				CIL 2 repurposed for leach duty	
	Feed Streams				Concentrate Pre-Leach Thickener Underflow & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Concentrate Pre-Leach Thickener Underflow & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Concentrate Pre-Leach Thickener Underflow & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Concentrate Pre-Leach Thickener Underflow & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	1	Owner	106		0
	Leach pH												
	30-TK-32A - 30-TK-32C				11.75 - 12.5	11.75 - 12.5	11.75 - 12.5	11.75 - 12.5	1	Owner	108		0
	30-TK-32D				11.5 - 12.5	11.5 - 12.5	11.5 - 12.5	11.5 - 12.5	1	Owner	108		0
	30-TK-32E				11 - 12.5	11 - 12.5	11 - 12.5	11 - 12.5	1	Owner	108		0
	Number of Tanks				7	7	7	7	1	Owner	114		1
	Effective Tank Capacity	Per Tank	m^3		1,685	1,685	1,685	1,685	1	Owner	16		A
	Total Tank Capacity		m^3		11,795	11,795	11,795	11,795	6	Calculated			
	Residence Time		h		119	86	86	107	6	Calculated			
	Minimum Residence Time		h		80	80	80	80	1	Owner	103		A
	Oxygen Addition Points				All Tanks	All Tanks	All Tanks	All Tanks	1	Owner	16		A
	Oxygen Addition Method				Side Lances on Tanks A to F and Oxygen Circulating Pump on Tanks A and B	Side Lances on Tanks A to F and Oxygen Circulating Pump on Tanks A and B	Side Lances on Tanks A to F and Oxygen Circulating Pump on Tanks A and B	Side Lances on Tanks A to F and Oxygen Circulating Pump on Tanks A and B	1	Owner	43		4
	Dissolved Oxygen Concentration	Target	ppm		10	10	10	10	1	Owner	16		A
	Cyanide Addition Points				All Tanks	All Tanks	All Tanks	All Tanks	1	Owner	43		A
	Cyanide Concentrations												
	30-TK-32A	Target	ppm		500	500	500	500	1	Owner	108		0
	30-TK-32B	Target	ppm		1,000	1,000	1,000	1,000	1	Owner	108		0
	30-TK-32C	Target	ppm		2,200	2,200	2,200	2,200	1	Owner	108		0
	30-TK-32D	Target	ppm		5,000	5,000	5,000	5,000	1	Owner	108		0
	30-TK-32E	Target	ppm		6,500	6,500	6,500	6,500	1	Owner	108		0
	Milk of Lime Addition Points				Leach Feed Box	Leach Feed Box	Leach Feed Box	Leach Feed Box	1	Owner	43		A
	Lead Nitrate Addition Points				Leach Feed Box	Leach Feed Box	Leach Feed Box	Leach Feed Box	1	Owner	43		A
10.3	**Adsorption (Brownfields)**				Existing CIL 3	Existing CIL 3	Existing CIL 3	Existing CIL 3				CIL 3 repurposed for adsorption duty	
	Number of Tanks				6	6	6	6	1	Owner	114		1
	Effective Tank Capacity	Per Tank	m^3		1,685	1,685	1,685	1,685	1	Owner	16		A
	Total Adsorption Tank Live Volume		m^3		10,110	10,110	10,110	10,110	6	Calculated			
	Residence Time		h		102	74	74	92	6	Calculated			
	Minimum Residence Time		h		60	60	60	60	1	Owner	103		A
	Total Leach & Adsorption Time		h		221	160	160	199	6	Calculated			
	Carbon Concentration	per tank	g/L		20	20	20	20	1	Owner	16		A
	Maximum Carbon Residence Time		days		14	14	14	14	1	Owner	103		A
	Carbon Movements per Day	Max	t/d		33	33	33	33	6	Calculated			
	Carbon In Circuit	Total	t		202.2	202.2	202.2	202.2	6	Calculated			
	Design Carbon Loading		g/t		4,490	4,490	4,490	4,490	1	Owner	16, 103		A
	Carbon Transfer Method				Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	1	Owner	16		A
	Carbon Transfer Period	per cycle	h		4.0	4.0	4.0	4.0	6	Calculated		Based on 11 t of Carbon & 20 g/L Carbon conc. within tanks	
	Carbon and Pulp Transfer Rate		m^3/h		138	138	138	138	2	EPC	119		4

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um					
10.4	**Loaded Carbon Recovery Screen & Transfer**												
	Loaded Carbon Advance Method				Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	2	EPC			1
	Loaded Carbon Pump & Pulp Forwarding Rate		m^3/h		275	275	275	275	2	EPC			1
	Loaded Carbon Transfer Time	Pump to Screen	h		2	2	2	2	2	EPC			1
	Feed Streams				Loaded Carbon Slurry from CIL3 Tank A (or Tank B)	Loaded Carbon Slurry from CIL3 Tank A (or Tank B)	Loaded Carbon Slurry from CIL3 Tank A (or Tank B)	Loaded Carbon Slurry from CIL3 Tank A (or Tank B)	1	Owner		As per P3-EPC-COR-602	1
	Type				Linear Motion	Linear Motion	Linear Motion	Linear Motion	1	Owner		New Derrick Linear Motion Screen as per P3-EPC-COR-602	1
	Number		No.		1	1	1	1	1	Owner		New Derrick Linear Motion Screen as per P3-EPC-COR-602	1
	Feed Rate		m^3/h		275	275	275	275	6	Calculated			
	Area		m^2		3.80	3.80	3.80	3.80	5	Vendor	117		2
	Deck Type				Derrick Polyweb Urethane	Derrick Polyweb Urethane	Derrick Polyweb Urethane	Derrick Polyweb Urethane	5	Vendor	117		1
	Aperture		mm		0.8 x 12.8	0.8 x 12.8	0.8 x 12.8	0.8 x 12.8	5	Vendor	117		1
	Total Spray Water Addition		m^3/h		10	10	10	10	3	Assumed			1
	Loaded Carbon Moisture		% w/w		40	40	40	40	1	Owner	4	Screen Oversize	1
	Loaded Carbon Transfer Moisture	Maximum	% w/w		55	55	55	55	1	Owner	4	From Transfer Hopper	1
	Loaded Carbon Transfer Hopper Capacity	Minimum	m^3		23	23	23	23	2	EPC			4
	Loaded Carbon Transfer Time	Hopper to Column	min		60	60	60	60	2	EPC			1
10.5	**Interstage Carbon Screens** (Brownfields)				Existing CIL 3	Existing CIL 3	Existing CIL 3	Existing CIL 3					
	Type				Mechanically Swept Pumped Screen	Mechanically Swept Pumped Screen	Mechanically Swept Pumped Screen	Mechanically Swept Pumped Screen	1	Owner	16		A
	Supplier & Model				Kemix MPSP 1200	Kemix MPSP 1200	Kemix MPSP 1200	Kemix MPSP 1200	1	Owner	16		A
	Number	per tank	No.		1	1	1	1	1	Owner	16		A
	Screen Feed Rate - per Screen	w/o carbon advance	m^3/h		102	140	140	113	6	Calculated			
		w/ carbon advance	m^3/h		240	278	278	251	6	Calculated			
	Area Selected	Selected	m^2		12.0	12.0	12.0	12.0	1	Owner	16		A
	Specific Screen Flux Rate	w/ carbon advance	$m^3/m^2/h$		20.0	23.1	23.2	20.9	6	Calculated			
		Maximum	$m^3/m^2/h$		100	100	100	100	1	Owner	16		B
	Screen Material				Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	1	Owner	102		A
	Screen Aperture		mm		1.0	1.0	1.0	1.0	1	Owner	16		A
10.6	**CIL 3 Tails Hopper**												
	Feed Flowrate		m^3/h		106	144	144	118	6	Calculated			
	Discharge Destination				CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	1	Owner	43		0
	Cyanide Concentration		ppm		750	750	750	750	1	Owner	108		0
11.0	**FLOTATION TAILINGS THICKENING**												
11.1	**Flotation Tailings Thickener**											Owner Supplied	
	Feed Stream				Scavenger Flotation Tails Cleaner-Scavenger Flotation Tails	Scavenger Flotation Tails Cleaner-Scavenger Flotation Tails	Scavenger Flotation Tails Cleaner-Scavenger Flotation Tails	Scavenger Flotation Tails Cleaner-Scavenger Flotation Tails	1	Owner	16		A
	Solids Feed Rate		t/h		3,071	2,858	3,042	3,411	6	Calculated			
	Feed Flowrate - Excl. Diluted Flocculant		m^3/h		6,871	6,492	6,884	6,187	6	Calculated			
	Feed Solids Concentration - Excl. Diluted Flocculant		% w/w		33.0	32.6	32.7	38.9	6	Calculated			
	Underflow Solids Concentration	Nominal	% w/w		55.9	56.0	56.0	55.9	6	Calculated		Achieve 55 % solids concentration in flotation tails CIL circuit post dilution (i.e. total feed % solids)	A
	Underflow Solids Concentration Range		% w/w		50-62	50-62	50-62	50-62	1	Owner	23		A
	Testwork Flux Rate		$t/m^2/h$		1.50	1.50	1.50	1.50	1	Owner	23		A
	Operating Flux Rate Range		$t/m^2/h$		1.05 - 1.27	1.05 - 1.27	1.05 - 1.27	1.05 - 1.27	1	Owner	16		A
	Nominal Flux Rate		$t/m^2/h$		1.09	1.01	1.08	1.21	6	Calculated			
	Thickener Area Calculated		m^2		2,827	2,827	2,827	2,827	6	Calculated			
	Diameter Selected		m		60	60	60	60	1	Owner	16		A
	Thickener Underflow Destination(s)				CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	1	Owner	43		A
	Number of Underflow Discharge Pumps				2	2	2	2	1	Owner	16		B
	Pump Configuration				Duty/Standby	Duty/Standby	Duty/Standby	Duty/Standby	1	Owner	43		A
	Total Thickener Underflow		m^3/h		3,329	3,087	3,289	3,701	6	Calculated			
12.0	**CIL - FLOTATION TAILINGS**												
12.1	**General Circuit**				CIL 4	CIL 4	CIL 4	CIL 4					
	Feed Streams				Flotation Tailings Thickener U/F, CIL 3 Tailings & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Flotation Tailings Thickener U/F, CIL 3 Tailings & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Flotation Tailings Thickener U/F, CIL 3 Tailings & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Flotation Tailings Thickener U/F, CIL 3 Tailings & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	1	Owner	106		0
	Number of Trains				1	1	1	1	1	Owner	43		A
	Fresh Solids Feed Flow Rate		t/h		3,151	2,969	3,153	3,500	6	Calculated		Flotation Tailings Thickener U/F & CIL 3 Tailings	
	Total Solids Feed Flow Rate		t/h		3,164	2,982	3,166	3,515	6	Calculated			
	Fresh Slurry Feed Flow Rate		m^3/h		3,438	3,234	3,437	3,822	6	Calculated		Flotation Tailings Thickener U/F & CIL 3 Tailings	
	Total Slurry Feed Flow Rate		m^3/h		3,511	3,305	3,510	3,900	6	Calculated			
	Target Feed Pulp Density	Target	% w/w		55	55	55	55	1	Owner	16, 103	Post dilution (i.e. total feed % solids)	A
	Maximum Feed Pulp Density		% w/w		60	60	60	60	1	Owner	43	Post dilution (i.e. total feed % solids)	A
	Feed Pulp Density	Actual	% w/w		55	55	55	55	6	Calculated		Post dilution (i.e. total feed % solids)	
	Leach Gold Extraction		%		38.0	38.0	38.0	38.0	1	Owner	44		B
	Adsorption Efficiency		%		88.0	88.0	88.0	88.0	1	Owner	44	Range 84 - 92%	B
12.2	**Leaching**				CIL 4	CIL 4	CIL 4	CIL 4					
	Leach pH												
	F21-TK-411				8.8 - 9.3	8.8 - 9.3	8.8 - 9.3	8.8 - 9.3	1	Owner	108		0
	Number of Tanks	per train			2	2	2	2	1	Owner	43		A
	Live Tank Capacity		m^3		5,600	5,600	5,600	5,600	1	Owner	43		B
	Total Tank Capacity		m^3		11,200	11,200	11,200	11,200	6	Calculated			
	Residence Time		h		3.2	3.4	3.2	2.9	6	Calculated			
	Oxygen Addition Points				All Leach Tanks	All Leach Tanks	All Leach Tanks	All Leach Tanks	1	Owner	43		A
	Oxygen Addition Method				Side Lances and Circulating Pumps on both Leach Tanks	Side Lances and Circulating Pumps on both Leach Tanks	Side Lances and Circulating Pumps on both Leach Tanks	Side Lances and Circulating Pumps on both Leach Tanks	1	Owner	43		A
	Dissolved Oxygen Concentration	Target	ppm		15	15	15	15	1	Owner	108		0
	Cyanide Addition Points				Feed Box and Tank 2	Feed Box and Tank 2	Feed Box and Tank 2	Feed Box and Tank 2	1	Owner	43		B
	Cyanide Concentration												
	F21-TK-411		ppm		175	175	175	175	1	Owner	108		0
	Milk of Lime Addition Points				Leach Feed Box	Leach Feed Box	Leach Feed Box	Leach Feed Box	1	Owner	43		A
12.3	**Adsorption**				CIL 4	CIL 4	CIL 4	CIL 4					
	Number of Tanks	per train			6	6	6	6	1	Owner	43	Provision in the layout for a future, 7th adsorption tank	A
	Live Tank Capacity		m^3		5,600	5,600	5,600	5,600	1	Owner	43		B
	Total Adsorption Tank Live Volume		m^3		33,600	33,600	33,600	33,600	6	Calculated			
	Residence Time		h		9.6	10	10	9	6	Calculated			
	Total Leach & Adsorption Time		h		12.8	13.6	12.8	11.5	6	Calculated		2 Leaching & 6 Adsorption Tanks will not achieve the 14 hours total residence time. Client is aware and has advised to maintain 8 tanks with the 9th as a future.	
	Oxygen Addition Points				First Two Adsorption Tanks	First Two Adsorption Tanks	First Two Adsorption Tanks	First Two Adsorption Tanks	1	Owner	43		A
	Oxygen Addition Method				Side Lances	Side Lances	Side Lances	Side Lances	1	Owner	43		A
	Carbon Concentration - Duty	per tank	g/L		12.5	12.5	12.5	12.5	1	Owner	109		1
	Carbon Concentration Used for Carbon Transfer		g/L		12.5	12.5	12.5	12.5	1	Owner	109		1
	Carbon Movements per Day	Max	t/d		30	30	30	30	6	Calculated			
	Carbon Residence Time	Maximum	d		14	14	14	14	1	Owner	103		A
	Carbon In Circuit	@ 8 g/L	t		269	269	269	269	6	Calculated			
		@ 10 g/L	t		336	336	336	336	6	Calculated			
		@ 12.5 g/L	t		420	420	420	420	6	Calculated			
	Nominal Carbon Loading		g/t		461	461	461	461	1	Owner	16, 103		A
	Carbon Loading Range		g/t		332 - 579	332 - 579	332 - 579	332 - 579	1	Owner	16, 103		A
	Carbon Transfer Method				Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	2	EPC			0
	Carbon Transfer Period		h		5.5	5.5	5.5	5.5	6	Calculated			
	Carbon and Pulp Transfer Rate		m^3/h		220	220	220	220	2	EPC			0
	Loaded Carbon Advance Method				Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	2	EPC			0
	Loaded Carbon Pump & Pulp Forwarding Rate		m^3/h		280	280	280	280	2	EPC			0
	Loaded Carbon Transfer Time		h		4.3	4.3	4.3	4.3	6	Calculated			
12.4	**Loaded Carbon Recovery Screen**				CIL 4	CIL 4	CIL 4	CIL 4					
	Feed Streams				First Adsorption Tank	First Adsorption Tank	First Adsorption Tank	First Adsorption Tank	1	Owner	16		A
	Type				Horizontal Vibrating	Horizontal Vibrating	Horizontal Vibrating	Horizontal Vibrating	1	Owner	16		A
	Number		No.		1	1	1	1	1	Owner	16		A
	Feed Rate		m^3/h		280	280	280	280	6	Calculated			
	Selected Size	Width	m		1.83	1.83	1.83	1.83	5	Vendor	84		A
		Length	m		4.88	4.88	4.88	4.88	5	Vendor	84		A
	Area		m^2		8.93	8.93	8.93	8.93	6	Calculated			
	Specific Screen Flux Rate	Actual	$m^3/m^2/h$		31	31	31	31	6	Calculated		w/o screen spray water	
		Maximum	$m^3/m^2/h$		35	35	35	35	1	Owner	103		A
	Deck Type				Polyurethane non-pegging	Polyurethane non-pegging	Polyurethane non-pegging	Polyurethane non-pegging	1	Owner	103		A
	Aperture		mm		0.8 x 18.5	0.8 x 18.5	0.8 x 18.5	0.8 x 18.5	5	Vendor	84		1
	Total Spray Water Addition		m^3/h		13	13	13	13	5	Vendor			0
	Loaded Carbon Moisture		% w/w		40	40	40	40	1	Owner	4	Screen Oversize	A
	Loaded Carbon Transfer Moisture		% w/w		55	55	55	55	1	Owner	4	From Transfer Vessel	B
12.5	**Intertank Carbon Screens**				CIL 4	CIL 4	CIL 4	CIL 4					
	Type				Mechanically Swept Pumped Screen	Mechanically Swept Pumped Screen	Mechanically Swept Pumped Screen	Mechanically Swept Pumped Screen	1	Owner	103		A
	Supplier & Model				Kemix MPS2535(P)	Kemix MPS2535(P)	Kemix MPS2535(P)	Kemix MPS2535(P)	5	Vendor			0
	Number	per tank	No.		2	2	2	2	1	Owner	43, 102		A
	Peak Screen Feed Rate - per Screen	w/o carbon advance	m^3/h		1,756	1,653	1,755	1,950	6	Calculated			
		w/ carbon advance	m^3/h		1,866	1,763	1,865	2,060	6	Calculated			
	Area Selected	Selected	m^2		25	25	25	25	5	Vendor			0
	Specific Screen Flux Rate - per screen	w/o carbon advance	$m^3/m^2/h$		69.3	65.2	69.2	76.9	6	Calculated			
		w/ carbon advance	$m^3/m^2/h$		73.6	69.5	73.6	81.3	6	Calculated			
	Screen Material				Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	1	Owner	102		A
	Screen Aperture		mm		1.0	1.0	1.0	1.0	1	Owner	102		A

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um					
12.6	**Carbon Safety Screen**				Future	Future	Future	Future					
	Type				Horizontal Vibrating	Horizontal Vibrating	Horizontal Vibrating	Horizontal Vibrating	1	Owner	103		A
	Number		No.		2	2	2	2	1	Owner	43	Duty / Duty	A
	Screen Feed Rate - per Screen		m^3/h		1,760	1,656	1,759	1,954	6	Calculated			
	Selected Size	Width	m		4.27	4.27	4.27	4.27	5	Vendor	84		0
		Length	m		7.93	7.93	7.93	7.93	5	Vendor	84		0
	Area - per Screen		m^2		34	34	34	34	6	Calculated			
	Area - Total		m^2		68	68	68	68	6	Calculated			
	Specific Screen Flux Rate	Nominal	$m^3/m^2/h$		52	49	52	58	6	Calculated	104	Screen flux rate of 58 $m^3/m^2/h$ accepted 16/02/2024	
		Max Recommended	$m^3/m^2/h$		50	50	50	50	1	Owner	103		B
	Screen Deck Type				Polyurethane	Polyurethane	Polyurethane	Polyurethane	1	Owner	102, 103		A
	Aperture		mm		0.85 x 18.5	0.85 x 18.5	0.85 x 18.5	0.85 x 18.5	5	Vendor	84		0
	Screen Spray Water	per Screen	m^3/h		263	263	263	263	5	Vendor		TBC - Future Equipment - Screen Spray Water to be finalised	0
12.7	**Carbon Scout**												
	Sample Feed Streams				All Adsorption Tanks	All Adsorption Tanks	All Adsorption Tanks	All Adsorption Tanks	1	Owner	43		A
	Manufacturer				Gekko Systems	Gekko Systems	Gekko Systems	Gekko Systems	8	Industry			A
	Model				MK6	MK6	MK6	MK6	5	Vendor	85		A
	Water Type				Scheme Water	Scheme Water	Scheme Water	Scheme Water	5	Vendor	85		A
	Water Flowrate		L/h		200	200	200	200	5	Vendor	85		A
12.8	**Final Tails Tank**												
	Feed Streams				CIL 4 Tails, Waste from Acid Wash, Fine Carbon Waste	CIL 4 Tails, Waste from Acid Wash, Fine Carbon Waste	CIL 4 Tails, Waste from Acid Wash, Fine Carbon Waste	CIL 4 Tails, Waste from Acid Wash, Fine Carbon Waste	1	Owner	43		B
	Solids Feed Rate		t/h		3,164	2,982	3,166	3,515	6	Calculated			
	Volumetric Feed Rate		m^3/h		3,562	3,356	3,561	3,951	6	Calculated			
	Feed Solids Concentration		% w/w		54.5	54.5	54.5	54.6	6	Calculated			
	Residence Time	Actual	min		6.9	7.3	6.9	6.2	6	Calculated			
	Tank Capacity	Live Volume	m^3		407	407	407	407	2	EPC	89	Derived using the volume from above the top of agitator blade to the tank overflow	B
12.9	**Tailings System**				By Others	By Others	By Others	By Others					
	Number of Pumps				4	4	4	4	1	Owner	16	By Others	A
	Arrangement				Duty / Duty / Duty / Standby	Duty / Duty / Duty / Standby	Duty / Duty / Duty / Standby	Duty / Duty / Duty / Standby	1	Owner	16	By Others	A
12.10	**CIL4 Static Screen**												
	Feed Stream				Event Pond / Silt Trap Discharge	Event Pond / Silt Trap Discharge	Event Pond / Silt Trap Discharge	Event Pond / Silt Trap Discharge	2	EPC			0
	Type				Sieve Bend	Sieve Bend	Sieve Bend	Sieve Bend	2	EPC			C
	Number of.				1	1	1	1	10	TBC			C
	Aperture		mm		1mm Wedge	1mm Wedge	1mm Wedge	1mm Wedge	10	TBC			C
	Solids Flowrate - Total		t/h		50	50	50	50	2	EPC		200 m^3/h @ 20 %w/w	C
	Slurry Flowrate - Total		m^3/h		200	200	200	200	2	EPC		2000-F00-MEM-GE-10002_A - C1 - Client Approved	C
	Oversize Product Destination				CIL4 Static Screen Trash Box	CIL4 Static Screen Trash Box	CIL4 Static Screen Trash Box	CIL4 Static Screen Trash Box	2	EPC			4
13.0	**ELUTION - FLOTATION CONCENTRATE**												
13.1	**General (Brownfields)**				Existing Elution Circuit 1 / 2								
	Operating Schedule	Nominal	days/week		6				1	Owner	67		B
	Strips per Day	Circuit Capacity			1.5				1	Owner	16	Elution circuits can average 1.5 strips/column/day	A
	Strips per Week	Operating Range			8 - 24				1	Owner	16	Existing Operating Envelope	A
	Number of Elution Circuits				2				1	Owner	16		A
	Circuit Capacity	each	t		11				1	Owner	16		A
	Total Circuit Capacity	total	t		22				6	Calculated			
	Dry Carbon Bulk Density		t/m³		0.47				8	Industry			A
	Loaded Carbon Gold Grade		g/t		4,490				1	Owner	16, 103		A
	Barren Carbon Gold Grade		g/t		40				1	Owner	16	Low end of range used in Mass Balance	2
	Barren Eluate Grade	Operating Range	g/t		3 - 8				1	Owner	23		A
	Loaded Carbon Mercury Grade	Nominal	Hg ppm		124				1	Owner	16		B
	Barren Carbon Mercury Grade	Nominal	Hg ppm		79				1	Owner	16		B
	Regenerated Carbon Mercury Grade	Nominal	Hg ppm		4				1	Owner	16		B
13.2	**Carbon Rinse Screens (Brownfields)**				Existing								
	Feed Stream				CIL 3 Loaded Carbon Transfer Vessel				1	Owner	43		4
	Type				Decline Vibrating Screen				1	Owner	16		A
	Number	per elution circuit	No.		1				1	Owner	16		A
	Selected Size	Width	m		1.2				1	Owner	16		A
		Length	m		2.4				1	Owner	16		A
	Area		m^2		2.9				6	Calculated			
	% Mass Split to Oversize		%		100				2	EPC			B
	Oversize Moisture %		% w/w		20				1	Owner	57		B
13.3	**Acid Washing (Brownfields)**				Existing								
	Feed Stream				Carbon Rinse Screen Oversize				1	Owner	43		A
	Number of Columns	per elution circuit			1				1	Owner	16		A
	Acid Type				Hydrochloric Acid				1	Owner	16		A
	Acid Concentration		% w/v HCl		3.0				1	Owner	16		A
	Acid Injection Time		min		17				1	Owner	16		A
	Acid Wash Solution Flowrate		BV/h		2				6	Calculated			
	Acid Soak Time		min		60				1	Owner	16		A
	Acid Water Rinse Time		min		120				1	Owner	16		A
	Acid Rinse Bed Volumes		BV		4				1	Owner	16		B
	Acid Rinse Bed Volumes per hour		BV/h		2				6	Calculated			
	Waste Acid Solution Destination				Final Tails Tank				1	Owner	16		A
	Acid Wash Temperature		°C		Ambient				1	Owner	44		A
13.4	**Elution (Brownfields)**												
	Number of Columns	per elution circuit			1				1	Owner	16		A
	Elution Type				Split AARL				1	Owner	16		A
	Pre-Soak Solution Strength		% w/v NaCN		1.7				1	Owner	16		A
			% w/v NaOH		2				1	Owner	16		A
13.5	**Pregnant & Lean Eluate Tanks (Brownfields)**												
	Number of Pregnant Eluate Tanks				2				1	Owner	16		B
	Pregnant Eluate Tank Capacity		m^3		200				1	Owner	16		A
	Number of Lean Eluate Tanks				1				1	Owner	16		A
	Lean Eluate Tank Capacity		m^3		200				1	Owner	16		A
13.6	**Elution Electrowinning (Brownfields)**				Existing Circuits								
	Number of Circuits				2				1	Owner	16		A
	Number of Cells per Circuit	Circuit #1			2				1	Owner	16		A
		Circuit #2			3				1	Owner	16		A
	Cathode Dimensions		m x m		1.00 x 1.00				1	Owner	106		0
	Effective Cross Sectional Area	Each Cathode	m^2		1.00				6	Calculated			
	Cathode Type				Stainless Steel				1	Owner	16		A
	Cathode Wire Diameter		µm		152				1	Owner	16		A
	Number Cathodes per Cell		No.		18				1	Owner	106		0
	Rectifier Size	Circuit #1	A		2,550				1	Owner	106		0
		Circuit #2	A		2,550				1	Owner	106		0
	Electrowinning Time	Circuit #1	h		8				1	Owner	16		A
		Circuit #2	h		12				1	Owner	16		A
	Eluate Circulation Tank Volume		m^3		200				1	Owner	16		A
	Barren Eluate Tank Volume		m^3		200				1	Owner	16		A
	Electrowinning Recovery		%		98				3	Assumed			2
14.0	**ELUTION - FLOTATION TAILINGS**												
14.1	**General**				New								
	Operating Schedule		d/wk		6				1	Owner	8		A
	Strips per Day	Design			2				1	Owner	8		A
	Strips per Week				12				6	Calculated			
	Column Capacity		t		15				1	Owner	8, 43		A
	Number of Elution Circuits				1				1	Owner	8		A
	Number of Columns				1				1	Owner	8	Single column that completes both acid washing and elution duties	A
	Column Duty				Acid Wash and Elution				1	Owner	8		A
	Column Materials of Construction				SAF2205				2	EPC	8		A
	Required Strip Frequency		cycles/wk		12				6	Calculated			
	Elution Carbon		t/wk		180				6	Calculated			
	Dry Carbon Bulk Density		t/m³		0.47				8	Industry			A
	Carbon Bed Volume		m³		31.9				6	Calculated			
	Loaded Carbon Gold Grade	Range	g/t		332 - 579				1	Owner	8, 16, 103		B
	Barren Carbon Gold Grade	Nominal	g/t		46				1	Owner	8, 16, 103		A
		Range	g/t		15 - 60				1	Owner	8, 16, 103		B
	Barren Carbon Mercury Grade	Nominal	g/t		37				1	Owner	16		B
		Range	g/t		20 - 72				1	Owner	16		B
	Total Volume of Barren Eluate	per day	m³/day		319				6	Calculated		To CIL 2	
		per hour	m³/h		13.3				6	Calculated			
14.2	**Loaded Carbon Transfer**												
	Carbon Transfer Time		min		30				1	Owner	103		A
	Carbon Transfer Vessel Capacity		t		30				1	Owner	8, 43		A
	Loaded Carbon Moisture		% w/w		55				1	Owner	4		B
	Carbon Solids SG		t/m^3		2.20				8	Industry			B
	Transfer Volume	Carbon and Water	m³		25.2				6	Calculated		15t column capacity	
	Transfer Volume	Water Only	m³		8.3				6	Calculated		Excluding moisture inherent with loaded carbon	
	Transfer Flowrate	Carbon and Water	m^3/h		50.3				6	Calculated			
	Transfer Flowrate	Water Only	m^3/h		16.7				6	Calculated		Excluding moisture inherent with loaded carbon	
	Carbon Transfer Concentration		g/L		596				6	Calculated			
	Transfer Water Type				Scheme Water				2	EPC			0

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um					
14.3	**Acid Washing**												
	Acid Washes Per Week		No.			12			6	Calculated			
	Acid Wash Column Capacity		t			**15**			1	Owner	8	Combined Acid Wash & Elution column	A
	Acid Type					**Hydrochloric Acid**			1	Owner	103		A
	Acid Dosing Method					**Pumped**			1	Owner	103		A
	Acid Concentration		% w/v HCl			**3**			1	Owner	103		A
	Acid Injection Time		min			**17**			1	Owner	103		A
	Acid Soak Time		min			**60**			1	Owner	103		A
	Acid Water Rinse Time		min			**120**			1	Owner	103		A
	Total Acid Wash Solution Flowrate		BV/h			**2**			1	Owner	8		A
	Total Acid Wash Solution Flowrate		m^3/h			63.8			6	Calculated			
			BV			0.6			6	Calculated			
	Dilute Acid Flowrate		m^3/h			63.8			6	Calculated			
	Dilute Acid Solution Used		m^3			18.085			6	Calculated			
	Neat Acid Solution Used		m^3			1.5			6	Calculated			
	Waste Acid Solution Destination					**Final Tails Tank**			1	Owner	43		B
	Acid Wash Temperature		°C			**Ambient**			1	Owner	44		A
14.4	**Water Rinse**												
	Water Flowrate		m^3/h			63.8			6	Calculated		Based on 2 BV/h	
	Acid Water Rinse Time		min			**120**			1	Owner	103		A
	Volume Water Used		m^3			128			6	Calculated			
	No. Bed Volumes		BV			**4**			1	Owner	8, 103		
	Solution Destination					**Final Tails Tank**			1	Owner	43		B
	Acid Rinse Temperature		°C			**Ambient**			1	Owner	44		A
	Rinse Water Source					**Scheme Water**			1	Owner	103		A
14.5	**Elution**												
	Elution Type					**Split AARL**			1	Owner	43		A
	Required Elutions Per Week		No.			**12**			1	Owner	8, 43		B
	Elution Column Capacity		t			**15**			1	Owner	8, 43		A
	Carbon Bed Volume		m^3			31.9			6	Calculated			
	Elution Column Operating Pressure	Top of Column	kPa(g)			**340**			2	EPC		To prevent flashing (boiling) of solution inside column.	B
14.6	**Pre-treatment (with Lean Eluate)**												
	Cycle Time		min			**30**			2	EPC			A
	Solution Flowrate		BV/h			**2**			1	Owner	8		A
			m^3/h			63.8			6	Calculated			
	Temperature		°C			**95**			2	EPC			A
	No. of Bed Volumes					1.0			6	Calculated		5 BVs of Pregnant Eluate required by Owner (Source 8)	
	Pre-soak Solution Volume		m^3			31.9			6	Calculated			
	Column Discharge Destination					**Lean Eluate Tank**			8	Industry			B
	Solution Strength		% w/v NaCN			**1.7**			1	Owner	103		A
			% w/v NaOH			**2**			1	Owner	103		A
	Sodium Cyanide Consumption		kg/pre-treat			543			6	Calculated			
		Neat Solution	m^3/pre-treat			1.51			6	Calculated			
	Caustic Consumption		kg/pre-treat			638			6	Calculated			
		Neat Solution	m^3/pre-treat			0.85			6	Calculated			
	Lean Eluate Solution Make-up Per Mix		m^3/mix			29.5			6	Calculated			
14.7	**Elution (with Lean Eluate)**												
	Elution Cycle Time		min			**150**			2	EPC			A
	Elution Solution Flowrate		BV/h			**2**			1	Owner	8		A
			m^3/h			63.8			6	Calculated			
	Eluate Consumed		BV			5.0			6	Calculated		5 BVs of Pregnant Eluate required by Owner (Source 15)	
			m^3			159.6			6	Calculated			
	Elution Temperature		°C			**130**			2	EPC			A
	Column Discharge Destination					**CIL4 Eluate Circulation Tanks**			8	Industry			A
14.8	**Elution & Cooling (with Scheme Water)**												
	Cycle Time - Scheme Water (with heaters)		min			90			6	Calculated			
	Elution Temperature		°C			**130**			2	EPC			C
	Water Flowrate		BV/h			**2**			1	Owner	8		A
			m^3/h			63.8			6	Calculated			
	Water Consumed		BV			3.0			6	Calculated			
			m^3			95.7			6	Calculated			
	Cycle Time - Scheme Water (no heaters, cooling)		min			**60**			2	EPC			A
	Elution Cooldown Temperature	Target	°C			**<80**			2	EPC			A
	Water Flowrate		BV/h			**2**			1	Owner	8		A
			m^3/h			63.8			6	Calculated			
	Water Consumed		BV			2.0			6	Calculated			
			m^3			63.8			6	Calculated			
	Total Cycle Time		min			**150**			2	EPC			A
	Water Flowrate		BV/h			**2**			1	Owner	8		A
			m^3/h			63.8			6	Calculated			
	Total Water Consumed		BV			5.0			6	Calculated		Total of 5 BVs of Lean Eluate required by Owner (Source 8)	
			m^3			159.6			6	Calculated			
	Column Discharge Destination					**CIL4 Eluate Circulation Tanks and Lean Eluate Tank**			8	Industry		When the CIL4 eluate circulation tank is full, discharge switches to the lean eluate tank	A
	Carbon Transfer In		min			30.0			6	Calculated			
			h			0.5			6	Calculated			
	Total Acid Time (including rinsing)		min			197			6	Calculated			
			h			3.3			6	Calculated			
	Total Elution Time (including Cooling)		min			330			6	Calculated			
			h			5.5			6	Calculated			
	Carbon Transfer Out		min			53.5			6	Calculated			
			h			0.9			6	Calculated			
	Total		h			10.2			6	Calculated			
14.9	**Heating**												
	Elution Heater Type					**LPG Fired Elution Heater**			1	Owner	43		A
14.10	**Lean Eluate Tank**												
	Total Volume of Elution Eluate Per Strip	Pregnant	m^3			160			6	Calculated			
	Number of Lean Eluate Tanks					**1**			1	Owner	43		A
	Total Live Volume Lean Eluate Tank	Minimum	m^3			160			6	Calculated			
14.11	**Elution Water Tank**												
	Number of Elution Water Tanks					**1**			1	Owner	43		B
	Total Live Volume Elution Water Tank		m^3			160			6	Calculated			
14.12	**Eluate Expansion Tank**												
	Number of Eluate Expansion Tanks					**1**			2	EPC			B
	Total Live Volume Eluate Expansion Tank		m^3			3.5			6	Calculated			
14.13	**Total Water Consumption Per Strip**												
	Carbon Transfer	Scheme	m^3/strip			8			6	Calculated			
	Acid Wash	Scheme	m^3/strip			17			6	Calculated			
	Water Rinse	Scheme	m^3/strip			128			6	Calculated			
	Water Elution & Cooling	Scheme	m^3/strip			160			6	Calculated			
	Carbon Eduction (Elution to Regen)	Scheme	m^3/strip			60			6	Calculated			
	Carbon Quench to Sizing Screen	Scheme	m^3/strip			60			6	Calculated			
	Total Per Strip	Scheme	m^3/strip			432			6	Calculated			
14.14	**Elution Electrowinning**												
	Number Circuits					**1**			2	EPC			A
	Number of Cells Per Circuit					**2**			5	Vendor	48	2 off Split (2 in 1) Clean-in-Place cells	A
	Cell Configuration					**Parallel - 2 off Split (2 in 1)**			5	Vendor	48		B
	Cell Size		m x m			**1.00 x 1.00**			5	Vendor	48		A
	Effective Cross Sectional Area		m^2			1.00			6	Calculated			
	Pregnant Eluate Volume		m^3			160			6	Calculated			
	Pregnant Eluate Grade		g Au/m^3			52.6			6	Calculated			
	Number of Eluate Volume Passes					**5.64**			5	Vendor	48		A
	Electrowinning Feed Flowrate per Circuit	Total	m^3/h			90.0			6	Calculated			
	Electrowinning Feed Flowrate per Split Cell	per Split Cell	m^3/h			45.0			6	Calculated			
	Electrowinning Feed Flowrate per Individual Cell	per Individual Cell	m^3/h			22.5			6	Calculated			
	Maximum Cell Linear Velocity		m/min			**0.450**			2	EPC	48		A
	Actual Cell Linear Velocity		m/min			**0.375**			5	Vendor	48		A
	Electrowinning Time		h			**10**			1	Owner	103		A
	Cathode Type					**Stainless Steel 316 Mesh**			5	Vendor	48		A
	Number Cathodes per Individual Cell		No.			**11**			5	Vendor	48		A
	Number Cathodes per Split Cell		No.			**22**			5	Vendor	48		A
	Number of Rectifiers		No.			**4**			5	Vendor	48		B
	Rectifier Size		A			**2,500**			5	Vendor	48		B
	Target Barren Solution Grade		ppm Au			**<5**			2	EPC	8, 103		A
	Barren Solution Pumping Time To Leach		h			**8**			2	EPC			B
	Cleaning Method					**Clean in Place**			1	Owner	8		A
15.0	**GOLDROOM**												
15.1	**Goldroom Operation** (Brownfields)					**Existing**							
	Feed Sources					**Flotation Con EW product (CIL2/3) & Flotation Tails EW product (CIL 4)**			1	Owner	43		A
	Sludge Filter Operating Days		days/week			**2**			1	Owner	16		A
	Sludge Dewatering Cone Capacity		m^3			**1.3**			1	Owner	16		A
	Sludge Filter Make					**TEFSA**			1	Owner	16		A
	Sludge Filter Model					**HPLE**			1	Owner	16	470x470mm plate filter, 6 bar design pressure	A
	Mercury Retort Oven Operating Days		days/week			**2**			1	Owner	16		A
	Mercury Retort Oven Make					**Locchead**			1	Owner	16		A
	Mercury Retort Oven Model					**F66OV66**			1	Owner	16	20 ft^3 electric powered mercury retort	A
	Calcine per week		kg			**800 - 1,200**			1	Owner	16	Existing, Calcine logs, Max Capacity 1,600 kg per cycle	A
	Smelting Furnace Operating Days		days/week			**2**			1	Owner	16		A
	Furnace Make					**Furnace Engineering**			1	Owner	16		A
	Furnace Model					**Westofen KBU 80/H**			1	Owner	16	2000 L x 3000 D x 1710 H, Weishaupt G3-1a burner	A
	Furnace Extraction Fan					**75 kW / 1,480 RPM**			1	Owner	16	IP55, 415V with 315x5SPB motor pulley and 212x5SPB fan pulley	A
	Furnace Scrubber					**Metalcraft SN50247/1**			1	Owner	16		A
	Smelts per week					**6**			1	Owner	16	Two downloads per week, three pours per day	A

AREA	PARAMETER		UNIT	DATA	REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
				1	2	3	4						
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um					
16.0	**CARBON REGENERATION - FLOTATION CONCENTRATE**												
16.1	**Barren Carbon Transfer** (Brownfields)				**Existing**								
	Transfer Water Source				**Carbon Transfer Water (Scheme Water)**				1	Owner	16, 103		A
	Carbon Transfer Time		min		**30**				1	Owner	16, 103		A
	Carbon Transfer Concentration		% w/w		**20**				1	Owner	57		B
	Transfer Water Flowrate	Each	m^3/h		88				6	Calculated		Based off 11t elution column	
	Transfer Water Required	Each	m^3		44				6	Calculated		Based off 11t elution column	
16.2	**Regeneration Kiln Dewatering Screens** (Brownfields)				**Existing**								
	Type				**Horizontal Vibrating Screen**				1	Owner	16, 103		A
	Number		No.		**3**				1	Owner	16, 103	1x Per Regeneration Kiln	A
	Specific Screen Flux Rate	Max Recommended	$t/m^2/h$		**4**				1	Owner	16, 103		A
	Selected Size	Width	m		**1.2**				1	Owner	16, 103		A
		Length	m		**2.4**				1	Owner	16, 103		A
	Area		m^2		2.9				6	Calculated			
	Feed % Solids		% w/w		**20**				1	Owner	57		B
	% Mass Split to Oversize		%		**100**				2	EPC			B
	Oversize Moisture %		% w/w		**60**				1	Owner	57		B
16.3	**Carbon Regeneration Kiln** (Brownfields)				**Existing**								
	Regeneration Kiln Type				**Horizontal**				1	Owner	16, 103		A
	Regeneration Kiln Model				**Metso RK600/650HR**				1	Owner	16, 103		A
	Number				**3**				1	Owner	16, 103		A
	Eluted Carbon to be Regenerated		%		**100**				1	Owner	16, 103		A
	Regenerated Carbon per Batch		t		11				6	Calculated			
	Temperature		°C		**750**				1	Owner	16, 103		A
	Required Kiln Operating time		h/batch		**20**				1	Owner	100		A
	Capacity - Nominal		kg/h		550				6	Calculated			
	Capacity - Design		kg/h		**650**				1	Owner	100		A
16.4	**Carbon Quench Tank** (Brownfields)				**Existing**								
	Carbon Quench Tank Volume	Each	m^3		**12**				1	Owner	16, 103	3 off Quench Tanks	A
	Carbon Transfer Time	Each	min		**30**				1	Owner	16, 103		A
	Carbon Transfer Concentration		% w/w		**20**				1	Owner	57		B
	Transfer Water Flowrate	Each	m^3/h		88				6	Calculated		Based off 11t elution column	
	Transfer Water				**Carbon Transfer Water (Scheme Water)**				1	Owner	16, 103		B
	Transfer Water Required	Each	m^3		44				6	Calculated			
16.5	**CIL 3 Barren Carbon Sizing Screen**				**New**								
	Type				**Horizontal Vibrating Screen**				1	Owner	100		A
	Number		No.		**1**				1	Owner	100		A
	Feed Rate		m^3/h		98				6	Calculated		Based off 1 x 11t elution columns	
	Feed Rate		t/h		22				6	Calculated		Based off 1 x 11t elution columns	
	Selected Size	Width	m		**1.53**				5	Vendor	84		0
		Length	m		**3.66**				5	Vendor	84		0
	Area		m^2		5.60				6	Calculated			
	Specific Screen Flux Rate	Actual	$m^3/m^2/h$		17.5				6	Calculated		w/o screen spray water	
	Deck Type				**Polyurethane**				1	Owner	100		A
	Aperture		mm		**1.25 x 18.5**				5	Vendor	84		1
	% Mass Split to Oversize		%		**100**				2	EPC	84		B
	Oversize Moisture %		% w/w		**40**				1	Owner	4		B
	Screen Spray Water		m^3/h		**13**				5	Vendor			0
16.6	**Regeneration Kiln Off-Gas Treatment** (Brownfields)												
	Venturi Scrubber				**Existing**								
	Number				**1**				1	Owner	16, 103		A
	Design Dust Load		mg/Nm^3		**375**				1	Owner	16, 103		A
	Design Mercury		g/h		**546**				1	Owner	16, 103		A
	Design VOC		g/h		**831**				1	Owner	16, 103		A
	Design Gas Temperature		°C		**250**				1	Owner	16, 103		A
	Water Type				**Process**				1	Owner	16, 103		A
	Packed Bed Wet Scrubber				**Existing**								
	Number				**1**				1	Owner	16, 103		A
	Design Gas Flowrate		Nm^3/h		**6,481**				1	Owner	16, 103		A
	Design Gas Temperature		°C		**80**				1	Owner	16, 103		A
	Water Type				**Process**				1	Owner	16, 103		A
	Mist Eliminator				**Existing**								
	Number				**1**				1	Owner	16, 103		A
	Design Gas Flowrate		Nm^3/h		**3,916**				1	Owner	16, 103		A
	Design Gas Temperature		°C		**58**				1	Owner	16, 103		A
	Water Type				**Scheme**				1	Owner	16, 103		A
	Regenerative Thermal Oxidiser				**Existing**								
	Number				**1**				1	Owner	16, 103		A
	Design Gas Flowrate		Nm^3/h		**4,132**				1	Owner	16, 103		A
	Design Gas Temperature		°C		**58**				1	Owner	16, 103		A
	Carbon Filters				**Existing**								
	Number				**2 (Duty / Standby)**				1	Owner	16, 103		A
	Design Gas Flowrate		Nm^3/h		**4,503**				1	Owner	16, 103		A
	Design Gas Temperature		°C		**80**				1	Owner	16, 103		A
	Design Mercury Loading		g/h		**387**				1	Owner	16, 103		A
	Stack Emission Limits												
	Mercury		ppmv		**0.20**				1	Owner	16, 103	FIM ERP Commissioning Report	A
	VOC		ppmv		**3.2**				1	Owner	16, 103		A
17.0	**CARBON REGENERATION - FLOTATION TAILINGS**												
17.1	**Barren Carbon Transfer**				**New - Flotation Tailings**								
	Transfer Water Source				**Carbon Transfer Water (Scheme Water)**				1	Owner	103		B
	Carbon Transfer Time		min		**53**				2	EPC			0
	Carbon Transfer Concentration		%w/w		**20**				1	Owner	57		B
	Transfer Water Flowrate		m^3/h		67				6	Calculated		Based off 1x 15t carbon batch	
	Transfer Water Required		m^3		60				6	Calculated		Based off 1x 15t carbon batch	
17.2	**Regeneration Kiln Dewatering Screen**				**New**								
	Type				**Horizontal Vibrating Screen**				5	Vendor	84		A
	Number		No.		**1**				1	Owner	103	One elution column	A
	Feed Flow Rate		m^3/h		75				6	Calculated			
	Solids Feed Rate		t/h		17				6	Calculated			
	Selected Size	Width	m		**1.53**				5	Vendor	84		0
		Length	m		**3.66**				5	Vendor	84		0
	Area		m^2		5.6				6	Calculated			
	Specific Screen Flux Rate	Maximum	$t/m^2/h$		**4**				1	Owner	103		B
		Actual	$m^3/m^2/h$		13.4				6	Calculated			
		Actual	$t/m^2/h$		3.0				6	Calculated			
	Deck Type				**Polyurethane**				5	Vendor	84		B
	Aperture		mm		**1.2 x 18.5**				5	Vendor	84		0
	Feed % Solids		% w/w		**20**				1	Owner	57		B
	% Mass Split to Oversize		%		**100**				2	EPC			B
	Oversize Moisture %	Nominal	% w/w		**40**				2	EPC	52		B
		Maximum	% w/w		**60**				1	Owner	57		B
	Screen Spray Water		m^3/h		**13**				5	Vendor			0
17.3	**Carbon Regeneration Kiln**				**New**								
	Reactivation Kiln Type				**Horizontal - Indirect Fired, Electric**				1	Owner	103		B
	Number				**1**				1	Owner	103		A
	Regenerated Carbon per Batch		t		30				6	Calculated		2 strips per day from 15t elution column	
	Regen Kiln Feed Hopper Capacity	% of full column size	%		**200**				1	Owner	103		B
	Regen Kiln Feed Hopper Capacity		t		30				6	Calculated			
	Required Kiln Operating time	Nominal	h/batch		**20**				8	Industry			A
	Capacity - Nominal		kg/h		**1,500**				1	Owner	43		A
	Operating Temperature		°C		**650 - 750**				1	Owner	102		A
	Design Operating Temperature		°C		**750**				1	Owner	103		
	Fraction of Carbon Regenerated		%		**100**				1	Owner	103		
	Regenerated Carbon Mercury Grade	Nominal	Hg ppm		**1.3**				1	Owner	16		B
		Range	Hg ppm		**0 - 4**				1	Owner	16		B
17.4	**Carbon Quench Tank**				**New**								
	Carbon Quench Tank Volume	Minimum	m^3		**20**				1	Owner			A
	Carbon Transfer Time		min		**53**				2	EPC			0
	Carbon Transfer Concentration		% w/w		**20**				2	EPC			C
	Transfer Water Flowrate		m^3/h		67.3				6	Calculated		Based off one 15t elution column	
	Transfer Water				**Quench Hopper filled with Scheme Water** **Carbon Transfer completed with Raw Water**				1	Owner	100		0
	Transfer Water Required		m^3		60				6	Calculated		Based off one 15t elution column	
17.5	**CIL 4 Carbon Sizing Screen**				**New**								
	Type				**Horizontal Vibrating Screen**				1	Owner	100		A
	Number		No.		**1**				1	Owner	103	Single CIL4 Train	A
	Feed Flow Rate		m^3/h		75				6	Calculated			
	Solids Feed Rate		t/h		17				6	Calculated			
	Selected Size	Width	m		**1.53**				5	Vendor	84		0
		Length	m		**3.66**				5	Vendor	84		0
	Area		m^2		5.6				6	Calculated			
	Specific Screen Flux Rate	Maximum	$m^3/m^2/h$		**35.0**				1	Owner	103		B
		Actual	$t/m^2/h$		3.0				6	Calculated			
		Actual	$m^3/m^2/h$		13.4				6	Calculated		w/o screen spray water	
	Deck Type				**Polyurethane**				5	Vendor	84		B
	Aperture		mm		**1.25 x 18.5**				5	Vendor	84		1
	% Mass Split to Oversize		%		**100**				2	EPC	84		B
	Oversize Moisture %		% w/w		**40**				1	Owner	4		B
	Screen Spray Water		m^3/h		**13**				5	Vendor			0

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um					
17.6	**Regeneration Kiln Off-Gas Treatment**					New							
	Mercury Recovery	Target	%			>92			1	Owner	43		B
	Design Mercury	New - Range	g/t			20 - 72			1	Owner	16	Brownfields barren carbon mercury grade CIL2/3 20th - 80th Percentile (2017 - 2023).	B
		New - Design	g/h			108			6	Calculated		Based on the upper limit of brownfields barren carbon mercury grade. 72g/t for 1,500 kg carbon/h through regen kiln	
	VOC	New	g/h			639			6	Calculated		Based on brownfields design VOC of existing Venturi Scrubber. Assumes 3x existing Carbon Regen Kilns running at design throughput. Estimate only - stack emmission criteria used as basis of design	
	Gas Temperature	New - Nominal	°C			316			5	Vendor	52		B
		New - Design	°C			400			5	Vendor	52		1
	Gas Flow	New - Nominal	Nm^3/h			1,335			6	Calculated	52	Based on 40% moisture CIL4 Carbon Regeneration Kiln feed	B
		New - Design	Nm^3/h			3,003			6	Calculated	52	Based on 60% moisture CIL4 Carbon Regeneration Kiln feed	B
	Carbon Fines Knock-Out Box												
	Number					1			5	Vendor	52		A
	Dimensions		m			1.22 x 1.8			5	Vendor	52		1
	Water Type	Sprays				Scheme			5	Vendor	52		A
	Water Flowrate		m^3/h			8.39			3	Assumed		Sprays used for maintenance only. Not a continuous consumption.	4
	Kiln Off-Gas Condenser												
	Number					1			5	Vendor	52		A
	Cooler Type					Shell and Tube			5	Vendor	52		1
	Water Type					Cooling Water (Scheme)			5	Vendor	52		A
	Kiln Mist Eliminator					New							
	Number					1			5	Vendor	52		A
	Dimensions		m			1.22 x 1.8			5	Vendor	52		1
	Water Type	Sprays				Scheme			5	Vendor	52		A
	Water Flowrate		m^3/h			8.39			3	Assumed		Sprays used for maintenance only. Not a continuous consumption.	4
	Kiln Duct Heater												
	Number					1			5	Vendor	52		A
	Power		kW			7.5			5	Vendor	52		1
	Kiln Adsorption Vessel												
	Number					1			5	Vendor	52		A
	Type					SIC Bed - CGS Dual Banjo			5	Vendor	52		B
	Dimensions		m			1.98 x 2.7			5	Vendor	52		1
	Carbon Type & Size		mm			3mm Mersorb HT			5	Vendor	52		1
	Kiln Exhaust Fan												
	Number					1			5	Vendor	52		A
	Design Duty		Am^3/h			3,003			5	Vendor	52	At normal conditions of: 1 atm and 20 deg C	1
	Power		kW			12.5			5	Vendor	52		1
	Kiln Discharge Stack												
	Number					1			1	Owner	52		A
	Height		m			35			1	Owner	52		B
	Stack Emission Limits												
	Mercury	Maximum	ppmv			0.20			1	Owner	43	At standard temperature and pressure	A
			$micrograms/m^3$			1,800			1	Owner	52		B
	Mercury	Vendor System	ppmv			0.027			6	Calculated			A
	VOC	Maximum	ppmv			3.20			1	Owner	43		A
18.0	**CYANIDE DESTRUCT**												
	Feed Streams	1			Kaltails Decant	Kaltails Decant	Kaltails Decant	Kaltails Decant	1	Owner	35	TBC with updated drawings by others	2
		2			offsite rom Decant & FIM I Decant	offsite rom Decant & FIM I Decant	offsite rom Decant & FIM I Decant	offsite rom Decant & FIM I Decant	1	Owner	35	TBC with updated drawings by others	2
		3			Decant Dam #3 & #4	Decant Dam #3 & #4	Decant Dam #3 & #4	Decant Dam #3 & #4	1	Owner	35	TBC with updated drawings by others	2
	Decant Water Recovered from Tailings	% of Total Tailings Water	%		49	49	49	49	1	Owner	35		A
	Decant Water from Kaltails		m^3/h		728	728	728	728	1	Owner	44		B
	Total Feed Flow to Cyanide Water Tank		m^3/h		1,940	1,872	1,941	2,071	6	Calculated			
	Method				Hydrogen Peroxide	Hydrogen Peroxide	Hydrogen Peroxide	Hydrogen Peroxide	1	Owner	35		A
	Residual WAD Cyanide Target	Stage 2			<1.5	<1.5	<1.5	<1.5	1	Owner	35		A
	Dosing Points	Location			Inline Mixers to each Decant Water line (3 off)	Inline Mixers to each Decant Water line (3 off)	Inline Mixers to each Decant Water line (3 off)	Inline Mixers to each Decant Water line (3 off)	1	Owner	35		2
		Number			3	3	3	3	1	Owner	35		2
	Dosing Method				Duty/duty/duty/standby pumps	Duty/duty/duty/standby pumps	Duty/duty/duty/standby pumps	Duty/duty/duty/standby pumps	1	Owner	35	Standby pump acts as a lag pump to the duty operating pumps in the event of high peroxide demand	2
	Dosing Control Method				Online CN_{WAD} Analyser	Online CN_{WAD} Analyser	Online CN_{WAD} Analyser	Online CN_{WAD} Analyser	1	Owner	35		A
19.0	**REAGENTS**												
19.1	**Activator - Copper Sulphate**												
	Delivery Form				Bulk Liquid	Bulk Liquid	Bulk Liquid	Bulk Liquid	1	Owner	64		A
	Delivery Size per Trailer		L		24,000	24,000	24,000	24,000	5	Vendor	112		1
	Number of Trailers per delivery	Maximum			2	2	2	2	1	Owner	64		A
	Delivery Size		L		48,000	48,000	48,000	48,000	6	Calculated			
	Delivery System				Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	1	Owner	64		A
	Delivery Side				Central Rear	Central Rear	Central Rear	Central Rear	1	Owner	64		B
	Solution Mix Strength (as $CuSO_4.5H_2O$)		% w/v		25	25	25	25	1	Owner	16		A
	Solution Mix Strength (as $CuSO_4$)		% w/v		16	16	16	16	1	Owner	64		A
	Supply Solution SG		t/m^3		1.17	1.17	1.17	1.17	1	Owner	16		A
	Storage Volume	Minimum	# deliveries		2.5	2.5	2.5	2.5	1	Owner	83		B
	Number of Storage Tanks				1	1	1	1	1	Owner	16		A
	Live Volume per Tank		m^3		120	120	120	120	6	Calculated			B
	Residence Time	Actual	d		11	12	11	10	6	Calculated		Residence time calculated using nominal consumption rates.	
	Consumption												
	SAG Mill Discharge	Nominal	g/t fresh feed to Cyclone Feed Hopper		35	35	35	35	1	Owner	23, 75	Added at Cyclone Feed Hopper	A
	Cleaner Scavenger	Nominal	g/t fresh cleaner scav feed		10	10	10	10	1	Owner	23, 75		A
	SAG Mill Discharge	Design	g/t fresh feed to Cyclone Feed Hopper		70	70	70	70	6	Calculated		Added at Cyclone Feed Hopper. 100% Design Factor for pumping.	
	Cleaner Scavenger	Design	g/t fresh cleaner scav feed		20	20	20	20	6	Calculated		100% Design Factor for pumping.	
	SAG Mill Discharge	Nominal	L/h		440	414	440	489	6	Calculated			
	Cleaner Scavenger	Nominal	L/h		7.3	5.8	6.3	8.1	6	Calculated			
	SAG Mill Discharge	Design	L/h		881	829	881	978	6	Calculated		100% Design Factor for pumping.	
	Cleaner Scavenger	Design	L/h		14.7	11.6	12.7	16.3	6	Calculated		100% Design Factor for pumping.	
	Nominal Consumption	as $CuSO_4.5H_2O$	L/h		447	414	440	489	6	Calculated			
		as $CuSO_4.5H_2O$	m^3/h		0.45	0.41	0.44	0.49	6	Calculated			
	Design Consumption	as $CuSO_4.5H_2O$	L/h		896	841	894	995	6	Calculated			
		as $CuSO_4.5H_2O$	m^3/h		0.90	0.84	0.89	0.99	6	Calculated			
19.2	**Collector - PAX**											Potassium Amyl Xanthate	
	Delivery Form				Bulk Liquid	Bulk Liquid	Bulk Liquid	Bulk Liquid	1	Owner	64		A
	Delivery Size per Trailer		L		24,000	24,000	24,000	24,000	5	Vendor	112		1
	Number of Trailers per delivery	Maximum			2	2	2	2	1	Owner	64		A
	Delivery Size		L		48,000	48,000	48,000	48,000	6	Calculated			
	Delivery System				Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	1	Owner	16		A
	Delivery Side				Central Rear	Central Rear	Central Rear	Central Rear	1	Owner	64		B
	Solution Mix Strength		% w/v		20	20	20	20	1	Owner	64		A
	Supply Solution SG		t/m^3		1.095	1.095	1.095	1.095	1	Owner	23		A
	Number of Storage Tanks				1	1	1	1	1	Owner	16		A
	Live Volume per Tank		m^3		270	270	270	270	6	Calculated			B
	Residence Time	Actual	d		7	7	7	6	2	EPC	10	Residence time calculated using nominal consumption rates.	A
		Required	d		7	7	7	7	1	Owner	10, 16		A
	Consumption												
	Rougher Feed	Nominal	g/t fresh mill feed		40	40	40	40	1	Owner	16, 75		A
	Scavenger Feed	Nominal	g/t fresh mill feed		35	35	35	35	1	Owner	16, 75		A
	Cleaner Conditioning	Nominal	g/t fresh mill feed		16	16	16	16	1	Owner	16, 75		A
	Cleaner Scavenger	Nominal	g/t fresh mill feed		16	16	16	16	1	Owner	16, 75		A
	Rougher Feed	Nominal	L/h		629	592	629	699	6	Calculated			
	Scavenger Feed	Nominal	L/h		551	518	551	611	6	Calculated			
	Cleaner Conditioning	Nominal	L/h		252	237	252	280	6	Calculated			
	Cleaner Scavenger	Nominal	L/h		252	237	252	280	6	Calculated			
	Total Consumption	Nominal	L/h		1,683	1,585	1,683	1,869	6	Calculated			
		Nominal	m^3/h		1.68	1.58	1.68	1.87	6	Calculated			
	Consumption												
	Rougher Feed	Design	g/t fresh mill feed		80	80	80	80	6	Calculated		100% Design Factor for pumping	
	Scavenger Feed	Design	g/t fresh mill feed		70	70	70	70	6	Calculated		100% Design Factor for pumping	
	Cleaner Conditioning	Design	g/t fresh mill feed		32	32	32	32	6	Calculated		100% Design Factor for pumping	
	Cleaner Scavenger	Design	g/t fresh mill feed		32	32	32	32	6	Calculated		100% Design Factor for pumping	
	Rougher Feed	Design	L/h		1,258	1,185	1,258	1,398	6	Calculated		100% Design Factor for pumping	
	Scavenger Feed	Design	L/h		1,101	1,037	1,101	1,223	6	Calculated		100% Design Factor for pumping	
	Cleaner Conditioning	Design	L/h		503.3	474.0	503.3	559.1	6	Calculated		100% Design Factor for pumping	
	Cleaner Scavenger	Design	L/h		503.3	474.0	503.3	559.1	6	Calculated		100% Design Factor for pumping	
	Total Consumption	Design	L/h		3,366	3,170	3,366	3,739	6	Calculated			
		Design	m^3/h		3.37	3.17	3.37	3.74	6	Calculated			

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um					
19.3	**Frother - Luprofroth 168**												
	Delivery Form				Bulk Liquid	Bulk Liquid	Bulk Liquid	Bulk Liquid	1	Owner	64		A
	Delivery Size per Trailer		L		26,000	26,000	26,000	26,000	5	Vendor			0
	Number of Trailers per delivery	Maximum			1	1	1	1	1	Owner	16		A
	Delivery Size		L		26,000	26,000	26,000	26,000	6	Calculated			
	Delivery System				Pumped Transfer From Delivery Truck	Pumped Transfer From Delivery Truck	Pumped Transfer From Delivery Truck	Pumped Transfer From Delivery Truck	1	Owner	64		A
	Delivery Side				Passenger	Passenger	Passenger	Passenger	1	Owner	64		B
	Solution Mix Strength		% w/w		100	100	100	100	1	Owner	64		A
	Supply Solution SG		t/m^3		1.082	1.082	1.082	1.082	1	Owner	16		A
	Number of Storage Tanks				1	1	1	1	2	EPC			A
	Live Volume per Tank		m^3		60	60	60	60	1	Owner	16	Live volume at 1.5 x maximum delivery (40,000 L)	A
	Residence Time	Minimum	d		7	7	7	7	1	Owner	10		A
		Actual	d		29	30	29	26	6	Calculated		Residence time calculated using nominal consumption rates.	
	Consumption												
	Rougher Feed	Nominal	g/t fresh mill feed		12	12	12	12	1	Owner	16, 75		A
	Scavenger Feed	Nominal	g/t fresh mill feed		6	6	6	6	1	Owner	16, 75		A
	Cleaner Feed	Nominal	g/t fresh mill feed		6	6	6	6	1	Owner	44, 75		A
	Cleaner Scavenger	Nominal	g/t fresh mill feed		6	6	6	6	1	Owner	16, 75		A
	Rougher Feed	Nominal	L/h		34.9	32.9	34.9	38.8	6	Calculated			
	Scavenger Feed	Nominal	L/h		17.4	16.4	17.4	19.4	6	Calculated			
	Cleaner Feed	Nominal	L/h		17.4	16.4	17.4	19.4	6	Calculated			
	Cleaner Scavenger	Nominal	L/h		17.4	16.4	17.4	19.4	6	Calculated			
	Total Consumption	Nominal	L/h		87.2	82.1	87.2	96.9	6	Calculated			
			m^3/h		0.09	0.08	0.09	0.10	6	Calculated			
	Consumption												
	Rougher Feed	Design	g/t fresh mill feed		24	24	24	24	6	Calculated		100% Design Factor for pumping	
	Scavenger Feed	Design	g/t fresh mill feed		12	12	12	12	6	Calculated		100% Design Factor for pumping	
	Cleaner Feed	Design	g/t fresh mill feed		12	12	12	12	6	Calculated		100% Design Factor for pumping	
	Cleaner Scavenger	Design	g/t fresh mill feed		12	12	12	12	6	Calculated		100% Design Factor for pumping	
	Rougher Feed	Design	L/h		69.8	65.7	69.8	77.5	6	Calculated		100% Design Factor for pumping	
	Scavenger Feed	Design	L/h		34.9	32.9	34.9	38.8	6	Calculated		100% Design Factor for pumping	
	Cleaner Feed	Design	L/h		34.9	32.9	34.9	38.8	6	Calculated		100% Design Factor for pumping	
	Cleaner Scavenger	Design	L/h		34.9	32.9	34.9	38.8	6	Calculated		100% Design Factor for pumping	
	Total Consumption	Design	L/h		174.4	164.3	174.4	193.8	6	Calculated			
			m^3/h		0.17	0.16	0.17	0.19	6	Calculated			
19.4	**Flocculant**												
	Packaging Type				Bulk Powder	Bulk Powder	Bulk Powder	Bulk Powder	1	Owner	64		A
	Delivery Size per Trailer		kg		22,500	22,500	22,500	22,500	1	Owner	64		A
	Number of Trailers per delivery	Maximum			1	1	1	1	1	Owner	64		A
	Delivery Size		kg		22,500	22,500	22,500	22,500	6	Calculated			
	Delivery System				Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	1	Owner	64		C
	Delivery Side				Passenger	Passenger	Passenger	Passenger	1	Owner	64		B
	Storage Volume	Minimum	# deliveries		2	2	2	2	1	Owner	23		A
	Number of Mixing Tanks				1	1	1	1	1	Owner	16		A
	Mixing Tank Live Volume		m^3		40	40	40	40	5	Vendor	36		B
	Solution Mix Strength		% w/v		0.30	0.30	0.30	0.30	1	Owner	44		A
	Mixes per Day per Tank				14	14	15	16	6	Calculated		Based on nominal Stage 2 consumption	
	Flocculant Dilution Factor				10	10	10	10	1	Owner	103		B
	Number of Storage Tanks				1	1	1	1	5	Vendor	36		A
	Storage Tank Live Volume		m^3		450	450	450	450	5	Vendor	36	Vendor advice.	A
	Residence Time		h		19	19	18	17	6	Calculated		Based on nominal Stage 2 consumption	
	Consumption												
	Flotation Tailings Thickener	Nominal	g/t		20	20	20	20	1	Owner	23		A
		Nominal	kg/h		61.4	57.2	60.8	68.2	6	Calculated			
		Nominal	L/h		20,473	19,054	20,278	22,738	6	Calculated			
		Design	L/h		40,946	38,108	40,557	45,476	6	Calculated			
	Concentrate Pre-Leach Thickener	Nominal	g/t		70	70	70	70	1	Owner	23		A
		Nominal	kg/h		5.2	7.3	7.3	5.8	6	Calculated			
		Nominal	L/h		1,748	2,430	2,430	1,949	6	Calculated			
		Design	L/h		3,496	4,859	4,860	3,897	6	Calculated			
	Concentrate Thickener	Nominal	g/t		70	70	70	70	1	Owner	23		A
		Nominal	kg/h		5.2	7.3	7.3	5.8	6	Calculated			
		Nominal	L/h		1,748	2,430	2,430	1,948	6	Calculated			
		Design	L/h		3,496	4,859	4,860	3,897	6	Calculated			
	Total Nominal Consumption	Stage 2	g/h		71,909	71,742	75,417	79,908	6	Calculated			
		Stage 2	L/h		23,970	23,914	25,139	26,636	6	Calculated			
		Stage 2	m^3/h		24.0	23.9	25.1	26.6	6	Calculated			
		Stage 2	t/a		588	587	617	654	6	Calculated			
	Total Design Consumption	Stage 2	g/h		95,000	95,000	95,000	95,000	2	EPC		Based on 28.35Mtpa Total Consumption for Stage 2 when the Regrind Mill was still included. Regrind Mill was moved post tender phase.	A
		Stage 2	t/a		777	777	777	777	6	Calculated			
		Stage 2	m^3/h		31.7	31.7	31.7	31.7	6	Calculated		Neat Concentration	
		Stage 2	L/h		47,938	47,826	50,277	53,270	6	Calculated		100% Design Factor	
		Stage 2	m^3/h		47.9	47.8	50.3	53.3	6	Calculated		100% Design Factor	
	Silo Capacity	Actual	t		45	45	45	45	5	Vendor	36		A
			d		26	26	25	23	6	Calculated		Based on nominal Stage 2 consumption	
19.6	**pH Modifier - Quicklime**				New	New	New	New					
	Type				Powder	Powder	Powder	Powder	1	Owner	64		B
	Packaging Type				Bulk	Bulk	Bulk	Bulk	1	Owner	64		A
	Delivery Size per Trailer		kg		30,000	30,000	30,000	30,000	1	Owner	64		A
	Number of Trailers Per Delivery		#		3	3	3	3	1	Owner	64		A
	Delivery Size		kg		90,000	90,000	90,000	90,000	6	Calculated			
	Preferred Delivery Method				Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	1	Owner	64		C
	Bulk Density		t/m^3		0.95	0.95	0.95	0.95	1	Owner	16		A
	Solids SG				3.37	3.37	3.37	3.37	1	Owner	16		A
	Water SG				1.07	1.07	1.07	1.07	1	Owner	16	Site metallurgical accounting assumption	A
	Solution Mix Strength		% w/w		20	20	20	20	1	Owner	16	Cyclone overflow density	A
	Lime Slurry Density		t/m^3		1.24	1.24	1.24	1.24	6	Calculated			
	Feed Size												
	F100		mm		6.7	6.7	6.7	6.7	1	Owner	116		1
	F80		µm		1,150	1,150	1,150	1,150	1	Owner	116		1
	Product Size		µm		38	38	38	38	1	Owner	116		1
	Dry Consumption												
	Concentrate CIL - Nominal	concentrate basis	kg/t		51.2	51.2	51.2	51.2	1	Owner	23		A
	Concentrate CIL - Design	concentrate basis	kg/t		120.1	120.1	120.1	120.1	1	Owner	16		A
	Flotation Tailings CIL - Nominal	tailings CIL basis	kg/t		3.20	3.20	3.20	3.20	1	Owner	23		A
	Flotation Tailings CIL - Design	tailings CIL basis	kg/t		4.50	4.50	4.50	4.50	1	Owner	16		A
	Total Dry Consumption												
	Nominal	@ 93.4% Availability	t/h		13.9	14.8	15.4	15.5	6	Calculated			
	Nominal	@ 75% Lime Utilisation	t/h		17.3	18.5	19.2	19.3	6	Calculated		75% lime plant utilisation as per 2000-F72-TQY-PR-15033	
	Design	@ 93.4% Availability	t/h		23.2	25.9	26.7	25.8	6	Calculated			
	Design	@ 75% Lime Utilisation	t/h		28.9	32.2	33.2	32.1	6	Calculated		75% lime plant utilisation as per 2000-F72-TQY-PR-15033	
	Milk of Lime Consumption												
	Nominal		m^3/h		56.2	59.9	62.2	62.4	6	Calculated			
	Peak		m^3/h		93.5	104.4	107.7	104.0	6	Calculated			
	New Circuit												
	Storage Silo Capacity	per Silo	t		1,132	1,132	1,132	1,132	5	Vendor	118		1
	Number of Silos				2	2	2	2	5	Vendor	118		1
	Residence Time	Actual	d		7	6	6	6	6	Calculated		Based on nominal consumption	
	Lime Slaker Type				Mill	Mill	Mill	Mill	1	Owner	43		A
	Number of Slakers				2	2	2	2	5	Vendor	118		1
	Slaker Power		kW/mill		6 x 55	6 x 55	6 x 55	6 x 55	5	Vendor	118	6 x 55 kW motors per slaking mill	2
	Utilisation		%		75	75	75	75	1	Owner	116		A
	Mill Throughput	Design	t/hr/mill		17	17	17	17	5	Vendor	118		2
			t/hr		33	33	33	33	5	Vendor	118		2
	Make-up Water Type				Raw Water	Raw Water	Raw Water	Raw Water	1	Owner	103		A
	Milk of Lime Storage Tank Live Volume	Max	m^3		524	524	524	524	1	Owner	107	524 m3 live volume based on minimum tank level (above pump suction). Live volume can reduce to as low as 406 m3 based on vendor recommended minimum slurry level (agitator submerged to a height equal to half the diameter).	0
	Number of Storage Tanks				1	1	1	1	1	Owner	43		A
	Residence Time	Actual	h		9.3	8.8	8.4	8.4	6	Calculated			
	Ringmain Flow Factor				2.5	2.5	2.5	2.5	2	EPC		Applied to Peak flow consumption rates	C
	Lime Ringmain Pumps	Required	m^3/h		234	261	269	260	6	Calculated			
		Selected	m^3/h		300	300	300	300	2	EPC			C

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um					
19.7	**Hydrogen Peroxide**												
	Type				Liquid	Liquid	Liquid	Liquid	1	Owner	64		A
	Supply Strength		% w/v		70	70	70	70	1	Owner	64		B
	Supply Solution SG		t/m³		1.29	1.29	1.29	1.29	5	Vendor	111		1
	pH				4.3	4.3	4.3	4.3	1	Owner	44		B
	Delivery Form				Bulk, Isotainer	Bulk, Isotainer	Bulk, Isotainer	Bulk, Isotainer	1	Owner	64		B
	Delivery Size per Trailer		L		24,000	24,000	24,000	24,000	5	Vendor			B
	Number of Trailers Per Delivery		#		2	2	2	2	1	Owner	64		0
	Delivery Size		L		48,000	48,000	48,000	48,000	6	Calculated			
	Delivery System				Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	1	Owner	64		B
	Delivery Side				Central Top	Central Top	Central Top	Central Top	1	Owner	64		B
	No. Dosing Points				3	3	3	3					
	Consumption Rate - Nominal	each Inline Mixer	L/h		140	140	140	140	1	Owner	35	One dosing pump per mixer.	A
		Total	L/h		420	420	420	420	6	Calculated			
	Consumption Rate - Design	each Inline Mixer	L/h		280	280	280	280	6	Calculated			
		Total	L/h		840	840	840	840	6	Calculated			
	Number of Storage Tanks				1	1	1	1	2	EPC		316 SS Storage Tank	A
	Storage Tank Live Volume	Per Tank	m^3		84.35	84.35	84.35	84.35	2	EPC		2000-F72-TQY-ME-10322_1	1
		Total	m^3		84	84	84	84	6	Calculated			
	Hydrogen Peroxide Storage Tank Residence Time												
	Required Residence Time	Nominal	days		7	7	7	7	1	Owner	10		B
	Nominal Usage Rate		days		8	8	8	8	6	Calculated			
19.8	**Sodium Cyanide**				New	New	New	New					
	Delivery Form				Bulk Liquid	Bulk Liquid	Bulk Liquid	Bulk Liquid	1	Owner	64		B
	Delivery Size per Trailer		L		22,000	22,000	22,000	22,000	1	Owner	64		A
	Number of Trailers per delivery	Maximum			2	2	2	2	1	Owner	64		A
	Delivery Size		L		44,000	44,000	44,000	44,000	6	Calculated			
	Delivery System				Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	1	Owner	64		C
	Delivery Side				Central Top	Central Top	Central Top	Central Top	1	Owner	64, 83	One per Storage Tank (2 off)	B
	Solution Mix Strength		% w/w		29	29	29	29	1	Owner	64		4
	Supply Solution SG		t/m³		1.24	1.24	1.24	1.24	2	EPC		NaCN SG at 29%w/w solution (Source: SysCAD)	4
	Consumption												
	Flotation Concentrate CIL	Operating Range	kg/t CIL Feed		2.0 - 8.0	2.0 - 8.0	2.0 - 8.0	2.0 - 8.0	1	Owner	23, 44	Dosed to six tanks	A
		No. of Dose Points	#		6	6	6	6	1	Owner	23, 44, 103	Dosed to six tanks	B
		Nominal	kg/t CIL per tank		3.58	3.58	3.58	3.58	1	Owner	44	Dosed to six tanks. Total of 21.50 kg/t as per Owner provided data.	B
		Nominal	kg/t CIL Total		21.5	21.5	21.5	21.5	6	Calculated			
		Nominal	kg/d		38,658	53,733	53,740	43,093	6	Calculated			
	Flotation Concentrate CIL	Design	kg/t CIL per tank		10	10	10	10	1	Owner	23, 44	Dosed to six tanks	A
		Design	kg/t CIL Total		60	60	60	60	6	Calculated			
			kg/d		107,882	149,951	149,973	120,260	6	Calculated		of NaCN	
	Flotation Tailings CIL	Operating Range	kg/t CIL Feed		0.04-0.24	0.04-0.24	0.04-0.24	0.04-0.24	1	Owner	23		A
		No. of Dose Points	#		4	4	4	4	1	Owner	103	CIL4 Feed Distribution Box, Leach tank 1, 2 and Adsorption Tank 1	C
	Flotation Tailings CIL	Nominal	kg/t CIL Total		0.22	0.22	0.22	0.22	1	Owner	23	Dosing split between dosing points	B
		Nominal	kg/d		16,637	15,678	16,648	18,480	6	Calculated			
	Flotation Tailings CIL	Design	kg/t CIL Feed		0.33	0.33	0.33	0.33	1	Owner	16	Dosing split between dosing points	B
			kg/d		24,956	23,517	24,972	27,720	6	Calculated			
	Elution Circuit 1	CIL2/3	m^3/strip		1.2	1.2	1.2	1.2	1	Owner	16		A
		Min. instantaneous	m^3/h		2.4	2.4	2.4	2.4	6	Calculated		Assumed 30 min pre-treatment time	
	Elution Circuit 2	CIL2/3	m^3/strip		1.2	1.2	1.2	1.2	1	Owner	16		A
		Min. instantaneous	m^3/h		2.4	2.4	2.4	2.4	6	Calculated		Assumed 30 min pre-treatment time	
	Elution Circuit 3	CIL4	m^3/strip		1.51	1.51	1.51	1.51	6	Calculated			
		Min. instantaneous	m^3/h		3.03	3.03	3.03	3.03	6	Calculated			
	Elution Consumption	Nominal	kg/d		2,375	2,375	2,375	2,375	6	Calculated			
		Design	kg/d		4,750	4,750	4,750	4,750	6	Calculated		100% Design Factor	
	Intensive Leach		kg/batch		230	230	230	230	5	Vendor	71	Based on CS10000 Acacia unit	B
		Nominal	kg/d		230	230	230	230	6	Calculated			
		Design	kg/d		460	460	460	460	6	Calculated		100% Design Factor	
	Nominal Consumption		kg/d		57,900	72,016	72,993	64,178	6	Calculated			
			kg/h		2,413	3,001	3,041	2,674	6	Calculated			
		NaCN solution (29%w/w)	L/h		6,734	8,375	8,489	7,464	6	Calculated			
			kg/t plant feed		0.77	1.01	0.97	0.77	6	Calculated			
	Design Consumption		kg/d		138,048	178,678	180,155	153,190	6	Calculated			
			kg/h		5,752	7,445	7,506	6,383	6	Calculated			
		NaCN solution (29%w/w)	L/h		16,054	20,779	20,951	17,815	6	Calculated			
			kg/t plant feed		1.83	2.51	2.39	1.83	6	Calculated			
	New Circuit												
	Number of Storage Tanks				2	2	2	2	1	Owner	83		B
	Live Volume per Tank	Each Tank	m^3		1,250	1,250	1,250	1,250	1	Owner	83		B
	Residence Time	Actual	d		15.5	12.4	12.3	14.0	6	Calculated		Based on nominal consumption.	
	Ringmain Flow Factor				2.0	2.0	2.0	2.0	2	EPC		Applied to Design flow consumption rates	0
	Cyanide Ringmain Pumps	Required	m^3/h		32.1	41.6	41.9	35.6	6	Calculated			
		Selected	m^3/h		50	50	50	50	2	EPC			0
19.9	**Oxygen**				New							Owner Supplied Package	
	Flotation Tails CIL												
	Consumption Rate	Nominal	Nm^3/t mill feed		0.17				6	Calculated	91		B
	Flotation Concentrate CIL												
	Consumption Rate	Nominal	Nm^3/t conc		13				1	Owner	23		A
	Total Consumption												
		Nominal	t/d		53.2				6	Calculated	91		B
		Maximum	t/d		70.8				6	Calculated	91		B
	New Oxygen Plant											Owner Supplied Package	
	VPSA Capacity	Selected	t/d		70				1	Owner	47	Pure Oxygen Basis	A
	VPSA Capacity		Nm^3/h		2,042				6	Calculated		Pure Oxygen Basis, density of O2 = 1.428 kg/m3 at 0°C and 101.3kPa	
	O_2 Purity	Minimum	%		~~92.0~~ 90%				1	Owner	47		A
	New LOX Plant											Owner Supplied Package	
	Liquid Oxygen Capacity		t		60				1	Owner	47		A
	Liquid Oxygen Pressure		kPa		850				1	Owner	16	As per existing oxygen plant	A
19.10	**Lead Nitrate**				New	New	New	New					
	Delivery Form				Bulk Liquid	Bulk Liquid	Bulk Liquid	Bulk Liquid	1	Owner	64		A
	Delivery Size per Trailer		L		24,000	24,000	24,000	24,000	5	Vendor	112		1
	Number of Trailers per delivery	Maximum			2	2	2	2	1	Owner	64		A
	Delivery Size		L		48,000	48,000	48,000	48,000	6	Calculated			
	Preferred Delivery System				Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	1	Owner	64		A
	Delivery Side				Central Rear	Central Rear	Central Rear	Central Rear	1	Owner	64		B
	Solution Mix Strength		% w/v		40	40	40	40	1	Owner	64		A
	Supply Solution SG		t/m³		1.17	1.17	1.17	1.17	1	Owner	16		A
	Consumption												
	Concentrate Leaching	Nominal	g/t CIL Feed		10,000	10,000	10,000	10,000	1	Owner	23		A
			L/h		1,873	2,603	2,604	2,088	6	Calculated			
	Concentrate Leaching	Design	g/t CIL Feed		15,000	15,000	15,000	15,000	1	Owner	16	50% addition rate design margin	A
			L/h		2,809	3,905	3,906	3,132	6	Calculated			
	New Circuit												
	Number of Storage Tanks				1	1	1	1	1	Owner	16		A
	Live Volume per Tank		m³		283	283	283	283	1	Owner	83		B
	Residence Time		d		6	5	5	6	6	Calculated		Based on Nominal Consumption	
	Dosing Pumps	Design	L/h		5,000	5,000	5,000	5,000	2	EPC			B
19.11	**Sodium Hydroxide (Caustic)**				New								
	Delivery Form				Bulk Liquid				1	Owner	64		A
	Delivery Size per Trailer		L		20,000				1	Owner	64		A
	Number of Trailers per delivery	Maximum			3				1	Owner	64		A
	Delivery Size		L		60,000				6	Calculated			
	Preferred Delivery System				Pumped Transfer From Truck				1	Owner	64		B
	Delivery Side				Passenger				1	Owner	64		B
	Solution Mix Strength		% w/w		50				1	Owner	64		A
	Supply Solution SG		t/m³		1.50				1	Owner	16		A
	Consumption												
	Elution Circuit 1		m^3/strip		1.7				1	Owner	16	Existing	A
		Min. instantaneous	m^3/h		3.4				6	Calculated		Assumed 30 min pre-treatment time	
	Elution Circuit 2		m^3/strip		1.7				1	Owner	16		A
		Min. instantaneous	m^3/h		3.4				6	Calculated		Assumed 30 min pre-treatment time	
	Elution Circuit 3		m^3/strip		0.85				6	Calculated			
		Min. instantaneous	m^3/h		1.70				6	Calculated			
	Intensive Leach		kg/batch		29				5	Vendor	71	Based on CS10000 Acacia unit	B
			L/batch		39				6	Calculated			
	Consumption												
	Elution Circuit 1		kg/d		1,913				6	Calculated			
	Elution Circuit 2		kg/d		1,913				6	Calculated			
	Elution Circuit 3		kg/d		1,277				6	Calculated			
	Intensive Leach		kg/d		58				6	Calculated		Based on two batches per day	
	Calculated Consumption	Nominal	kg/d		5,160				6	Calculated			
		Nominal	kg/h		215				6	Calculated			
		Nominal	L/h		287				6	Calculated			
	New Circuit												
	Number of Storage Tanks				1				2	EPC			C
	Live Volume		m³		250				2	EPC			A
	Residence Time	Nominal	d		36				6	Calculated		Based on Nominal Consumption	
	Ringmain Flow Factor				2.5				2	EPC		Applied to combined instantaneous flowrate from Elution Circuits 1 & 2	0
	Caustic Ringmain Pumps	Required	m^3/h		17				6	Calculated			
		Selected	m^3/h		20				2	EPC			0

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um					
19.12	**Hydrochloric Acid**					**New**							
	Delivery Form					**Bulk Liquid**			1	Owner	64		A
	Delivery Size per Trailer		L			**20,000**			1	Owner	64		A
	Number of Trailers per delivery	Maximum				**3**			1	Owner	64		A
	Delivery Size		L			60,000			6	Calculated			
	Preferred Delivery System					**Pumped Transfer From Truck**			1	Owner	64		A
	Delivery Side					**Passenger**			1	Owner	64		B
	Supply Strength		% w/w			**32**			1	Owner	64		A
	Supply Solution SG		t/m³			**1.16**			1	Owner	16		A
	Consumption												
	Acid Wash Column 1		m³/strip			**2.7**			1	Owner	16		A
		Min. instantaneous	m³/h			9.5			6	Calculated			
	Acid Wash Column 2		m³/strip			**2.7**			1	Owner	16		A
		Min. instantaneous	m³/h			9.5			6	Calculated			
	Acid Wash Column CIL4		m³/strip			1.5			6	Calculated			
		Min. instantaneous	m³/h			5.2			6	Calculated			
	Consumption												
	Acid Wash Column 1		kg/d			1,503			6	Calculated			
	Acid Wash Column 2		kg/d			1,503			6	Calculated			
	Acid Wash Column CIL4		kg/d			1,085			6	Calculated			
	Calculated Consumption		kg/d			4,092			6	Calculated			
			kg/h			170			6	Calculated			
			L/h			256			6	Calculated		MB estimated, based on CIL4 acid injection time (17 min)	
			L/h			459			6	Calculated		Based on 2.7 m³ HCl / strip	
	New Circuit												
	Number of Storage Tanks					**1**			1	Owner	24		A
	Live Volume per Tank		m³			**200**			2	EPC			A
	Residence Time	Nominal	d			18			6	Calculated		Based on Nominal Consumption	
	Dilute Acid Storage Tank Live Volume		m³			**25**			2	EPC	110		0
	Neat Acid Dosing Pumps	Required	m³/h			**20**			2	EPC		2 off Elution Circuit 1/2 strips occuring simultaneously	C
	Dilute Acid Dosing Pump	Minimum	m³/h			**64**			2	EPC			0
19.13	**Grinding Media**												
	SAG Mill												
	Number of Bunkers				**2**	**2**	**2**	**2**	1	Owner	16		A
	Bunker Storage Size		m³		**35**	**35**	**35**	**35**	1	Owner	16		A
	Media Type				**Forged Steel**	**Forged Steel**	**Forged Steel**	**Forged Steel**	1	Owner	16, 38		A
	Media Diameter		mm		**125**	**125**	**125**	**125**	1	Owner	16, 38		A
	Bulk Density		t/m³		**4.65**	**4.65**	**4.65**	**4.65**	1	Owner	16		A
	Package Form				**Bulk**	**Bulk**	**Bulk**	**Bulk**	1	Owner	16		A
	Delivery Size		t		**25**	**25**	**25**	**25**	1	Owner	23		A
	Trailers per Delivery		#		**1**	**1**	**1**	**1**	1	Owner	23		A
	Consumption Rate - Existing SAG		kg/t mill feed		**0.35**	**0.35**	**0.35**	**0.35**	1	Owner	16, 38		A
	Consumption Rate - New SAG		kg/t mill feed		**0.40**	**0.40**	**0.40**	**0.40**	1	Owner	16, 38		A
	Total Consumption		t/a		9,759	10,649	10,724	10,899	6	Calculated			
	Storage Capacity		t		325.5	325.5	325.5	325.5	6	Calculated			
	Residence Time		d		12	11	11	11	6	Calculated			
	Indexer/Charger Storage Capacity	Minimum	Buckets		**1.5**	**1.5**	**1.5**	**1.5**	1	Owner	16	CAT972 Bucket	A
	Ball Mill												
	Number of Bunkers				**1**	**1**	**1**	**1**	1	Owner	16		A
	Bunker Storage Size		m³		**35**	**35**	**35**	**35**	1	Owner	16		A
	Media Type				**High Chrome**	**High Chrome**	**High Chrome**	**High Chrome**	1	Owner	16, 38		A
	Media Diameter		mm		**75**	**75**	**75**	**75**	1	Owner	16, 38		A
	Bulk Density		t/m³		**4.75**	**4.75**	**4.75**	**4.75**	1	Owner	16		A
	Package Form				**Bulk**	**Bulk**	**Bulk**	**Bulk**	1	Owner	16		A
	Delivery Size		t		**25**	**25**	**25**	**25**	1	Owner	23		A
	Trailers per Delivery		#		**1**	**1**	**1**	**1**	1	Owner	23		A
	Consumption Rate		kg/t mill feed		**0.20**	**0.20**	**0.20**	**0.20**	1	Owner	16, 38		A
	Consumption		t/a		5,148	5,699	5,699	5,718	6	Calculated			
	Storage Capacity		t		166.3	166.3	166.3	166.3	6	Calculated			
	Residence Time		d		12	11	11	11	6	Calculated			
	Indexer/Charger Storage Capacity	Minimum	Buckets		**1.5**	**1.5**	**1.5**	**1.5**	1	Owner	16	CAT972 Bucket	A
	UFG Mill												
	Media Type				**Ceramic**	**Ceramic**	**Ceramic**	**Ceramic**	1	Owner	16		A
	Media Diameter		mm		**2.5**	**2.5**	**2.5**	**2.5**	1	Owner	16		A
	Bulk Density		t/m³		**2.2**	**2.2**	**2.2**	**2.2**	1	Owner	16		A
	Package Form				**Bulk Bag**	**Bulk Bag**	**Bulk Bag**	**Bulk Bag**	1	Owner	16		A
	Package Weight		kg		**2,000**	**2,000**	**2,000**	**2,000**	1	Owner	16		A
	Addition Rate		kg/t mill feed		**0.56**	**0.56**	**0.56**	**0.56**	1	Owner	10		B
		UFG Mill 1	kg/h		**0.0**	**6.8**	**6.8**	**0.0**	6	Calculated			
		UFG Mill 1	kg/day		0.0	163.0	163.2	0.0	6	Calculated			
		UFG Mill 2	kg/h		**21.0**	**25.8**	**25.8**	**23.4**	6	Calculated			
		UFG Mill 2	kg/day		503.4	618.2	618.2	561.2	6	Calculated			
		UFG Mill 3	kg/h		**21.0**	**25.8**	**25.8**	**23.4**	6	Calculated			
		UFG Mill 3	kg/day		503.4	618.2	618.2	561.2	6	Calculated			
	Site Storage Capacity	Minimum	# Bags		**50**	**50**	**50**	**50**	1	Owner	10		B
		Design	# Bags		**54**	**54**	**54**	**54**	2	EPC		Navisworks model. Designed to achieve the minimum.	
			days		107	77	77	96	6	Calculated		UFG Mill 1, 2 & 3 Consumption	B
	Lime Slaking Mills												
	Media Type				**Steel**	**Steel**	**Steel**	**Steel**	5	Vendor	118		2
	Media Diameter		mm		**25.0**	**25.0**	**25.0**	**25.0**	5	Vendor	118		2
	Bulk Density		t/m³		**4.85**	**4.85**	**4.85**	**4.85**	5	Vendor	118		2
	Total Consumption		kg/mill/week		**170**	**170**	**170**	**170**	5	Vendor	118	Estimated. Dependant on lime slaking usage and hardness	2
			kg/week total		340	340	340	340	6	Calculated		Estimated. Dependant on lime slaking usage and hardness	
			kg/t		10.2	10.2	10.2	10.2	6	Calculated		Estimated using vendor provided mill ball consumption and lime design throughput	
	Package Form				**Bulk Bag**	**Bulk Bag**	**Bulk Bag**	**Bulk Bag**	1	Owner	118		2
	Media Hopper Capacity	UFG Mill 2	m³		**14**	**14**	**14**	**14**	5	Vendor	65	Isacharger capacity negligible.	B
	Residence Time	UFG Mill 2	days		61.2	49.8	49.8	54.9	6	Calculated			
	Residence Time	UFG Mill 2	# Bags		15.4	15.4	15.4	15.4	6	Calculated			
	Media Hopper Capacity	UFG Mill 3	m³		**14**	**14**	**14**	**14**	5	Vendor	65	Isacharger capacity negligible.	B
	Residence Time	UFG Mill 3	days		61.2	49.8	49.8	54.9	6	Calculated			
	Residence Time	UFG Mill 3	# Bags		15.4	15.4	15.4	15.4	6	Calculated			
19.14	**Antiscalant**												
	Type					**FF585**			1	Owner	16	Dosing pump for each dosing point	A
	Addition Rate - Gravity Concentrators		ml/min			**40**			1	Owner	16	New addition in Stage 1, to be dosed into gravity concentrator fluidisation water line	A
	Addition Rate - Kiln Scrubber (Existing)		ml/min			**45**			1	Owner	16	New addition in Stage 2	A
	Addition Rate - CIL4 Kiln Mercury Abatement System		ml/min			**61**			1	Owner	44	New addition in Stage 2	B
	Total Consumption		ml/min			146			6	Calculated			
	Number of IBC's for Storage		No.			2			6	Calculated		Based on 7 days storage & 1,000 L IBC volumes	
	Type					**FF540**			1	Owner	16	Dosing pump for each dosing point	A
	Addition Rate - Elution Circuit 1 (Existing)		ml/min			**56**			1	Owner	16	New addition in Stage 2	A
	Addition Rate - Elution Circuit 2 (Existing)		ml/min			**70**			1	Owner	16	New addition in Stage 2	A
	Addition Rate - Elution Circuit 1 Heat Exchanger (Existing)		ml/min			**24**			1	Owner	16	New addition in Stage 2	A
	Addition Rate - Elution Circuit 2 Heat Exchanger (Existing)		ml/min			**30**			1	Owner	16	New addition in Stage 2	A
	Addition Rate - Elution Circuit CIL4 (New)		ml/min			**95**			1	Owner	44	New addition in Stage 2	B
	Addition Rate - Elution Circuit CIL4 Heat Exchanger (New)		ml/min			**41**			1	Owner	44	New addition in Stage 2	B
	Total Consumption		ml/min			316			6	Calculated			
	Number of IBC's for Storage		No.			4			6	Calculated		Based on 7 days storage & 1,000 L IBC volumes	
19.15	**LPG**												
	Type					**Propane**			1	Owner	16		A
	Elution Burner 1 Range	High Flame	MJ/h			**4,900**			1	Owner	16	Existing equipment fed from upgraded LPG Storage Vessel	A
		Low Flame	MJ/h			**2,000**			1	Owner	16	Existing equipment fed from upgraded LPG Storage Vessel	A
	Elution Burner 2 Range	High Flame	MJ/h			**4,900**			1	Owner	16	Existing equipment fed from upgraded LPG Storage Vessel	A
		Low Flame	MJ/h			**2,000**			1	Owner	16	Existing equipment fed from upgraded LPG Storage Vessel	A
	CIL4 Elution Burner Range	Pre-Heat	L/h			**823**			5	Vendor	81	New addition in Stage 2 fed from upgraded LPG Storage Vessel	A
		Steady State	L/h			**708**			5	Vendor	81	New addition in Stage 2 fed from upgraded LPG Storage Vessel	A
	Regeneration Kiln 1		MJ/h			**4,500**			1	Owner	16	Existing equipment fed from upgraded LPG Storage Vessel	A
	Regeneration Kiln 2		MJ/h			**4,500**			1	Owner	16	Existing equipment fed from upgraded LPG Storage Vessel	A
	Regeneration Kiln 3		MJ/h			**4,500**			1	Owner	16	Existing equipment fed from upgraded LPG Storage Vessel	A
	Regenerative Thermal Oxidiser		MJ/h			**1,580**			1	Owner	88	Existing equipment fed from upgraded LPG Storage Vessel	B
	Smelting Furnace	Maximum	MJ/h			**2,000**			1	Owner	16	Existing equipment fed from dedicated Goldroom LPG Storage Vessels	A
	Consumption Rate	Plant Feed Basis	L/t			**0.40**			6	Calculated	50	Inclusive of Regeneration Kiln 3 and RTO but not the Goldroom Smelting Furnace. Calculated from a 25.5 Mtpa basis.	B
	Number of LPG Storage Vessels (Upgraded)					**1**			1	Owner	88	Client supply	B
	Vessel Capacity		kL			**161**			1	Owner	88	Client supply	B
	Residence Time	Actual	d			5.3			6	Calculated		As per source 82, Owner confirm and accept they won't achieve the 7 days	

AREA	PARAMETER		UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
					1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um					
20.0	**WATER SERVICES - EXISTING**												
20.1	**Saline Bore Water**												
	Source					Borefields			1	Owner	16		A
	Storage Method 1					Dam 1 (North)			1	Owner	16		A
	Live Volume		m^3			8,000			1	Owner	16		A
	Dam Water Pumps		#			2			1	Owner	23		A
	Pump Model					Stalker 250-250-400, 132kW			1	Owner	23		A
	Storage Method 2					Dam 2 (South)			1	Owner	16		A
	Live Volume		m^3			12,000			1	Owner	16		A
	Dam Water Pumps		#			3			1	Owner	106		0
	Pump Model					TKL 250x250 - 400HT			1	Owner	23		A
	Total Live Volume		m^3			20,000			6	Calculated			
20.2	**Mill Circuit Water**												
	Source				Concentrate Pre-Leach Thickener O/F Concentrate Thickener O/F	Concentrate Pre-Leach Thickener O/F Concentrate Thickener O/F	Concentrate Pre-Leach Thickener O/F Concentrate Thickener O/F	Concentrate Pre-Leach Thickener O/F Concentrate Thickener O/F	1	Owner			B
	Storage Method				Circuit Water Tank	Circuit Water Tank	Circuit Water Tank	Circuit Water Tank	1	Owner	16		A
	Live Volume		m^3		1,500	1,500	1,500	1,500	1	Owner	16		A
	Residence Time		min		100	94	94	100	6	Calculated			
	Water Demand - Nominal	Calculated	m^3/h		896	958	958	899	6	Calculated			
	Water Demand - Design		m^3/h		1,800	1,800	1,800	1,800	1	Owner	16		A
	Design Flowrate per Pump		m^3/h		1,800	1,800	1,800	1,800	1	Owner	16		A
	Pump Configuration				Duty / Standby	Duty / Standby	Duty / Standby	Duty / Standby	1	Owner	16		A
	Pump Model				APP51-300	APP51-300	APP51-300	APP51-300	1	Owner	23		A
20.3	**Cyanide Water**												
	Source					Decant Return			1	Owner	16		A
	Storage Method					Cyanide Water Tank			1	Owner	16		A
	Live Volume		m^3			1,115			1	Owner	16		A
	Cyanide Water Pump Configuration					Duty / Standby			1	Owner	16		A
	Cyanide Water Pump Model					TKL 200x200 - 500HT			1	Owner	23		A
	Cyanide Transfer Pump Configuration					Duty / Standby			1	Owner	16		A
	Cyanide Transfer Pump Model					TKL 100x65 - 315HT			1	Owner	23		A
20.4	**Scheme Water**												
	Water Demand		m^3/h		172	172	173	175	6	Calculated			
	Scheme Water Quality												
	SG		-		1.00	1.00	1.00	1.00	1	Owner	1		B
	TDS		mg/L		454	454	454	454	1	Owner	1		B
	pH		-		8.0	8.0	8.0	8.0	1	Owner	1		B
	Chlorides	Range	mg/L		232	232	232	232	1	Owner	1		B
20.5	**Fire Water / Safety Shower / Potable Water**					By Others			1	Owner	69	By Others	B
20.6	**Gland Water**												
	Source					Dam 2 (South)			1	Owner	16		A
	Storage Method					Gland Water Tank			1	Owner	16		A
	Live Volume		m^3			83			1	Owner	16		A
	Pump Configuration					Duty / Standby			1	Owner	16		A
	Pump Model					TIK 90kW			1	Owner	23		A
20.7	**Eastern Borefield Transfer Dam**												
	Source					Seepage			1	Owner	16		A
	Storage Method					Dam			1	Owner	16		A
	Destination					Dam 3			1	Owner	79		B
	Live Volume		m^3			12,000			1	Owner	16		A
	Pump Configuration					Duty / Standby			1	Owner	16		A
20.8	**Saline Gravity Dam**												
	Source					Borefield			1	Owner	16		A
	Storage Method					Dam			1	Owner	16		A
	Destination					Dam 1 (North)			1	Owner	79		B
	Live Volume		m^3			16,000			1	Owner	16		A
20.9	**Kaltails Saline Dam**												
	Source					Borefield			1	Owner	16		A
	Storage Method					Dam			1	Owner	16		A
	Destination					Dam 3			1	Owner	79		B
	Live Volume		m^3			13,600			1	Owner	16		A
	Pump Configuration					Duty / Duty / Standby			1	Owner	16		A
21.0	offsite ref **Saline Transfer Dam**												
	Source					Borefield			1	Owner	16		A
	Storage Method					Dam			1	Owner	16		A
	Destination					Dam 1 (North) and Dam 2 (South)			1	Owner	79		B
	Live Volume		m^3			340			1	Owner	16		A
	Pump Configuration					Duty / Standby			1	Owner	16		A
20.10	**Southern Surge Tank**												
	Storage Method					Southern Surge Tank			1	Owner	16		A
	Destination					Dam 2 (South)			1	Owner	16		A
	Capacity		m^3			575			1	Owner	16		A
21.0	**WATER SERVICES - NEW**												
21.1	**Process Water**												
	Source				Flotation Tailings Thickener O/F Saline Bore Water Makeup	Flotation Tailings Thickener O/F Saline Bore Water Makeup	Flotation Tailings Thickener O/F Saline Bore Water Makeup	Flotation Tailings Thickener O/F Regrind Thickener O/F Saline Bore Water Makeup	1	Owner	10		A
	Storage Method				Tank	Tank	Tank	Tank	9	Agreed	10		A
	Live Volume		m^3		2,080	2,080	2,080	2,080	9	Agreed	10		A
	Residence Time	Minimum	min		15	15	15	15	1	Owner	16		A
	Water Demand	Calculated	m^3/h		5,355	5,152	5,489	4,591	6	Calculated			
	Pump Configuration				Duty / Standby	Duty / Standby	Duty / Standby	Duty / Standby	9	Agreed	10		A
	Process Water Quality												
	SG		-		1.07	1.07	1.07	1.07	1	Owner	1		B
	TDS		mg/L		140,000	140,000	140,000	140,000	1	Owner	1		B
	pH		-		4.6	4.6	4.6	4.6	1	Owner	1		B
	Chlorides	Range	mg/L		20,492 - 86,243	20,492 - 86,243	20,492 - 86,243	20,492 - 86,243	1	Owner	1		B
21.2	**Raw Water**												
	Source				Dam 2 (South)	Dam 2 (South)	Dam 2 (South)	Dam 2 (South)	1	Owner	43	Supplies the Raw Water Tank	A
	Storage Method				Steel Tank Separate Raw, UFG 2/3 Gland, LP and HP Gland	Steel Tank Separate Raw, UFG 2/3 Gland, LP and HP Gland	Steel Tank Separate Raw, UFG 2/3 Gland, LP and HP Gland	Steel Tank Separate Raw, UFG 2/3 Gland, LP and HP Gland	1	Owner	68		B
	Configuration								1	Owner	68		B
	Total Live Volume	Actual	m^3		375	375	375	375	1	Owner	68		B
	Total Live Volume	Required	m^3		377	377	377	377	6	Calculated			
	Residence Time	Actual	min		60	60	60	60	6	Calculated			
	Residence Time	Required	min		60	60	60	60	9	Agreed	10, 16	Nominal Basis	B
	Total Nominal Water Demand (water consumption)		m^3/h		377	377	377	377	2	EPC	68	Note: Pump design for instantaneous flows in accordance with mechanical calculation.	B
	Pump Configuration				Duty / Standby	Duty / Standby	Duty / Standby	Duty / Standby	1	Owner	16		A
	Source (Greenfields)				Dam 3	Dam 3	Dam 3	Dam 3	1	Owner	69	By Others	B
	Live Volume		m^3		35,000	35,000	35,000	35,000	1	Owner	114	By Others	1
21.3	**Gland Water**												
	Source				Raw Water Tank	Raw Water Tank	Raw Water Tank	Raw Water Tank	1	Owner	43		A
	Gland Water Demand	Design	m^3/h		92	92	92	92	2	EPC			0
	Low Pressure Gland Water Pump Configuration				Duty / Standby	Duty / Standby	Duty / Standby	Duty / Standby	1	Owner	43		A
	High Pressure Gland Water Pump Configuration				Duty / Standby	Duty / Standby	Duty / Standby	Duty / Standby	1	Owner	43		A
22.0	**AIR SERVICES**												
22.1	**Low Pressure Flotation Air**												
	Number of Flotation Blowers					4			1	Owner	49		A
	Operating Configuration					Duty / Duty / Duty / Standby			1	Owner	49		A
	Type					Low pressure, oil free screw blower with sound enclosure			1	Owner	49		A
	Capacity - Each	Design	FAD m^3/h			8,688			1	Owner	49		A
	Capacity - Total		FAD m^3/h			26,064			6	Calculated			
	Pressure	Design	kPa(g)			119.7			1	Owner	49		A
22.2	**High Pressure Air**												
	Number of Compressors					3			1	Owner	43		A
	Operating Configuration					Duty / Duty / Standby			1	Owner	43		A
	Type					Oil Injected Rotary Screw Compressor			5	Vendor	45		B
	Capacity - Each	NTP Conditions	L/s			750			5	Vendor	45		A
	Pressure		kPa(g)			980			5	Vendor	45		B
	Duty					New facilities only			2	EPC			A